常見伴侶動物神經科疾病復健

國立臺灣大學附設動物醫院

復健及整合醫學科

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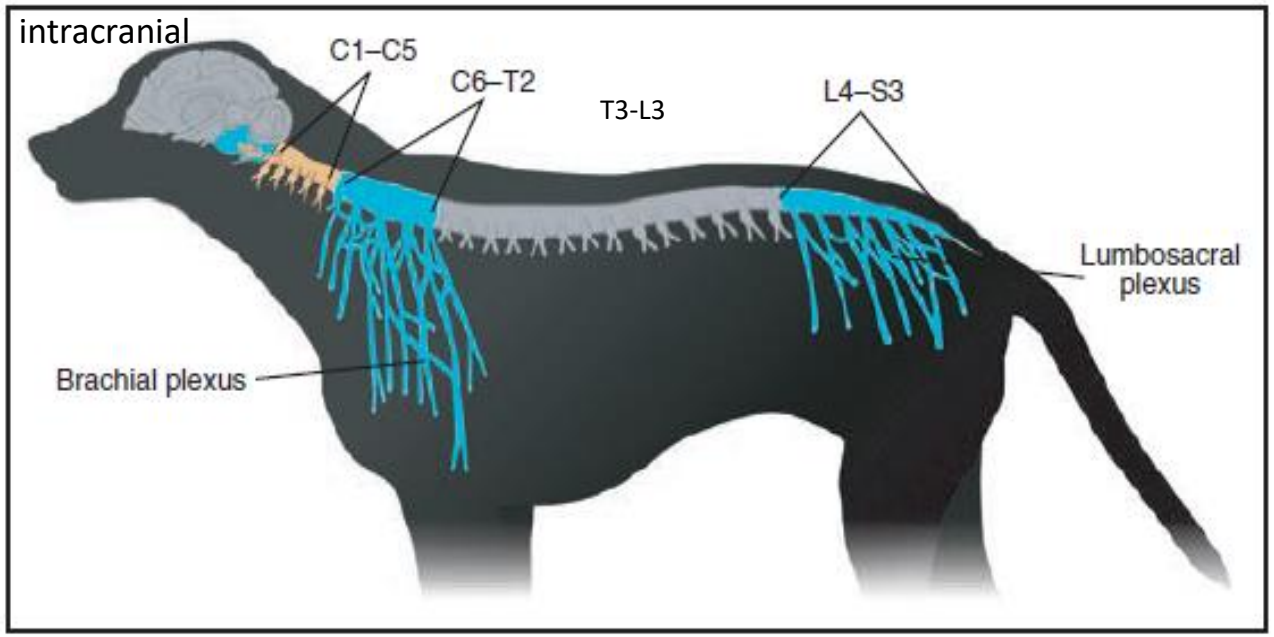
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Rehabilitation for common neurological diseases

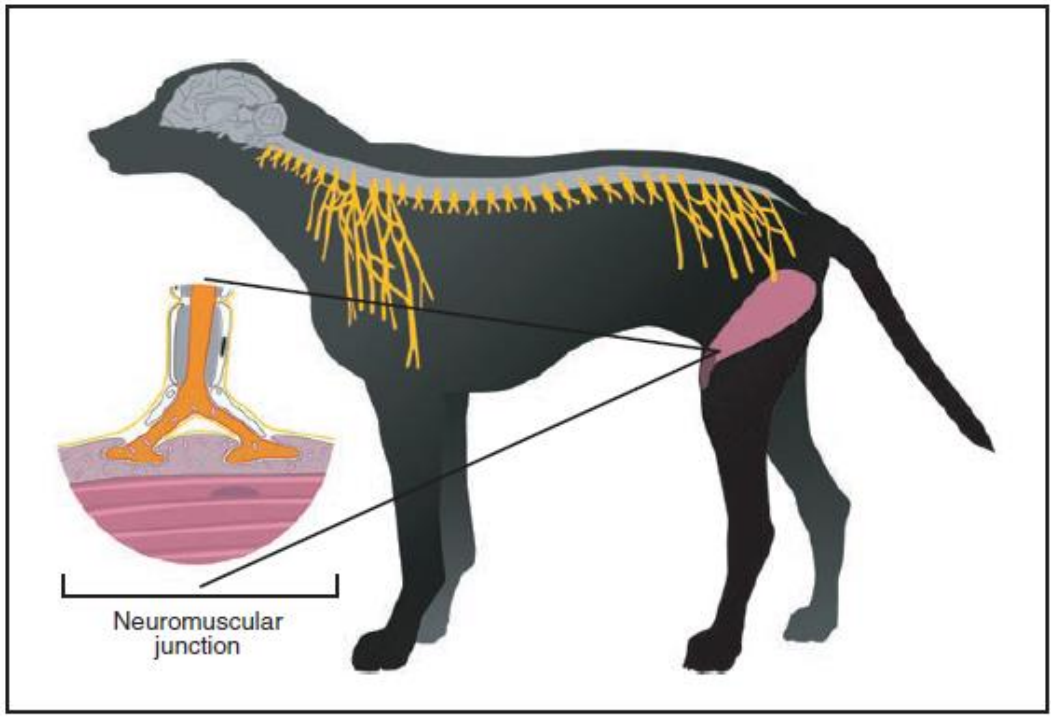
- Anatomy解剖學
- Lesion localization病灶定位
- Common neurological diseases for rehab 復健常見神經科疾病

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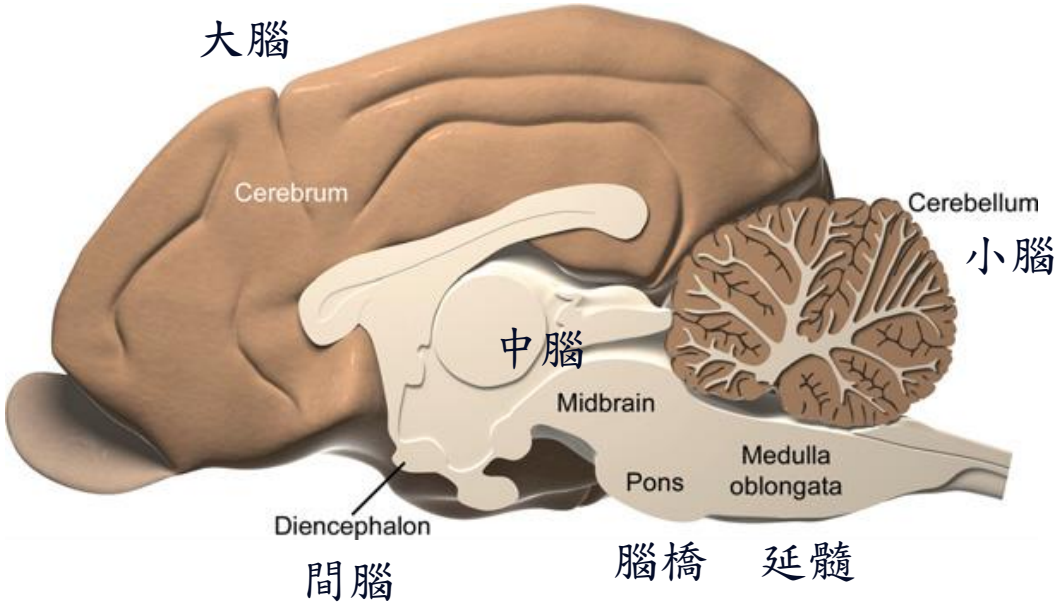


BSAVA Manual of Canine and Feline Neurology, 4th Edition, Olby 2014



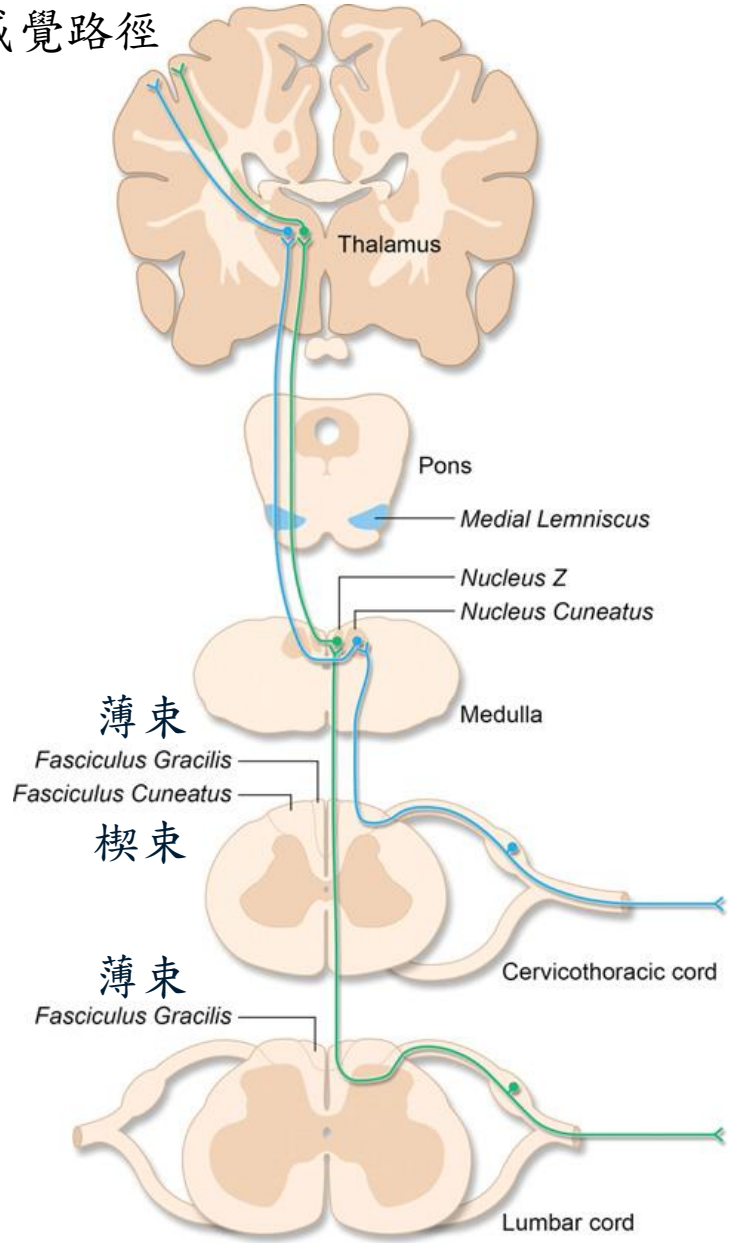
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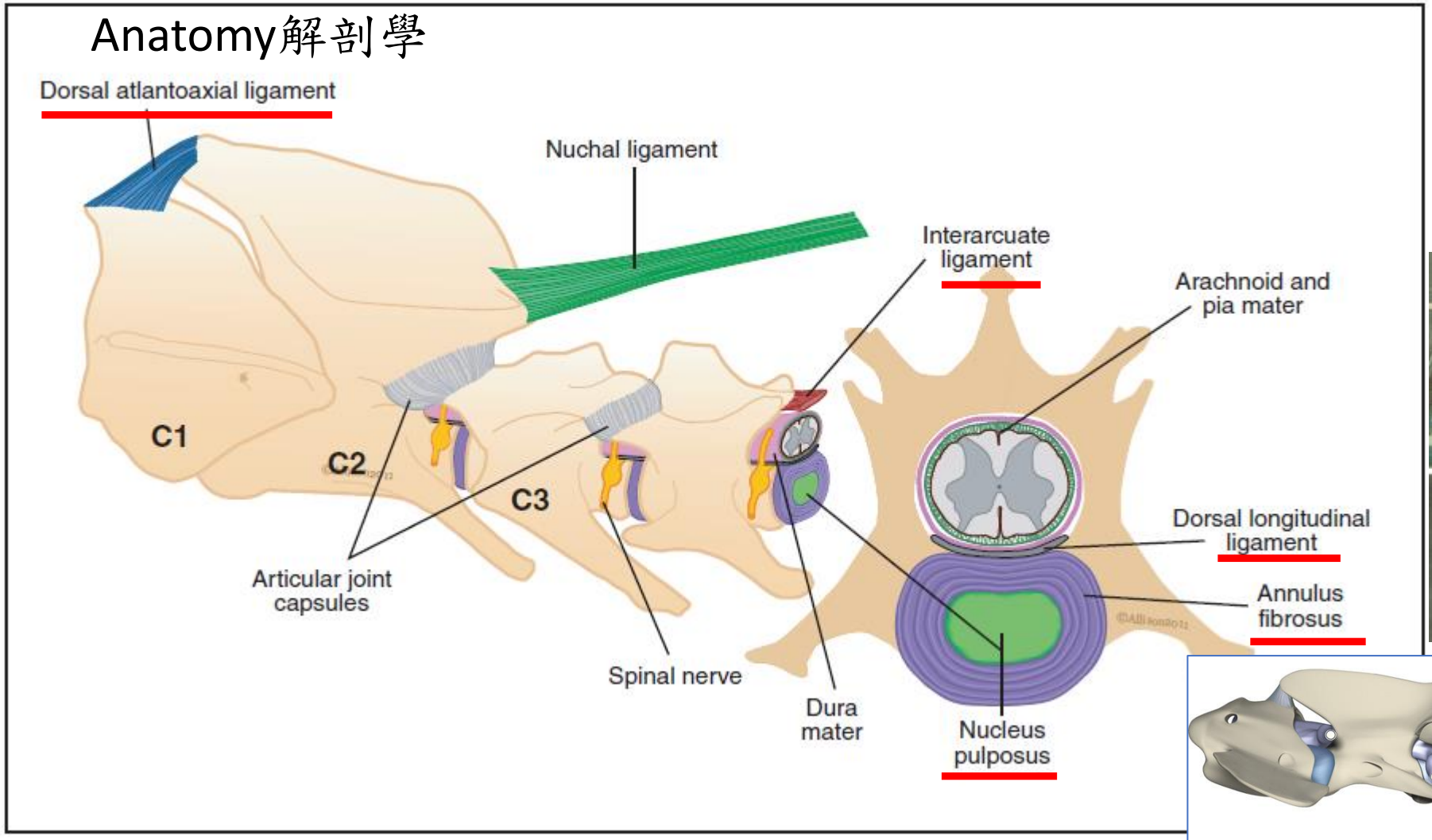
Conscious proprioceptive pathways

本體感覺路徑

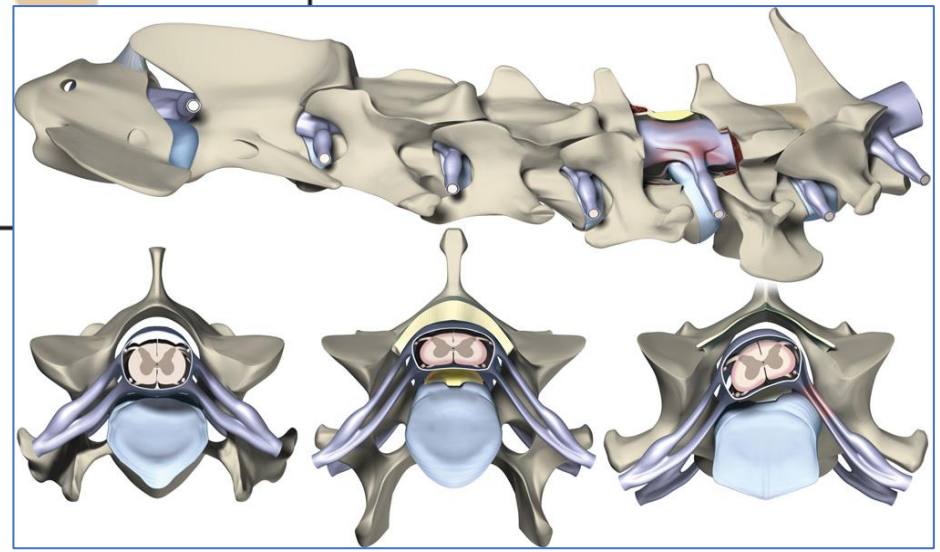


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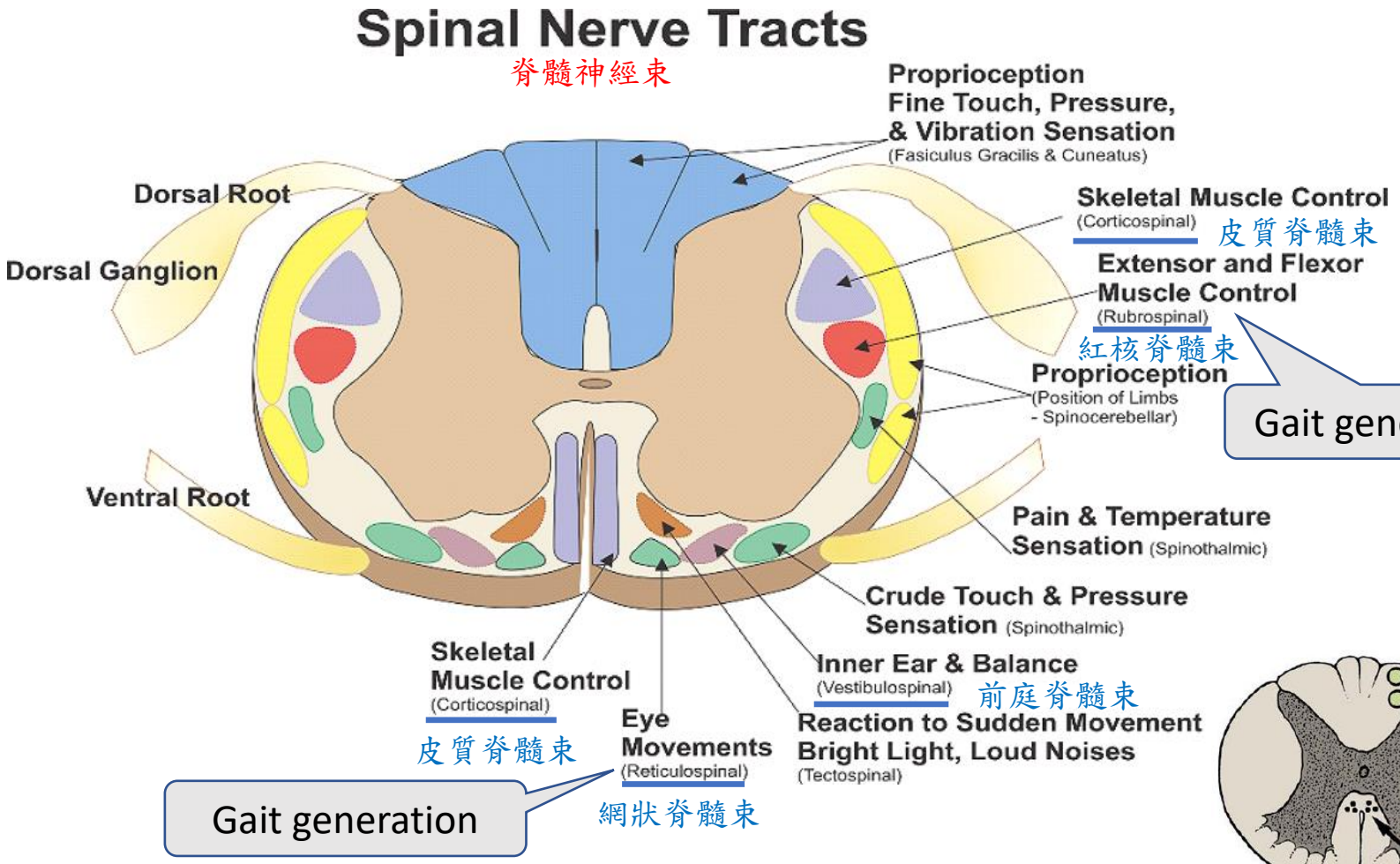


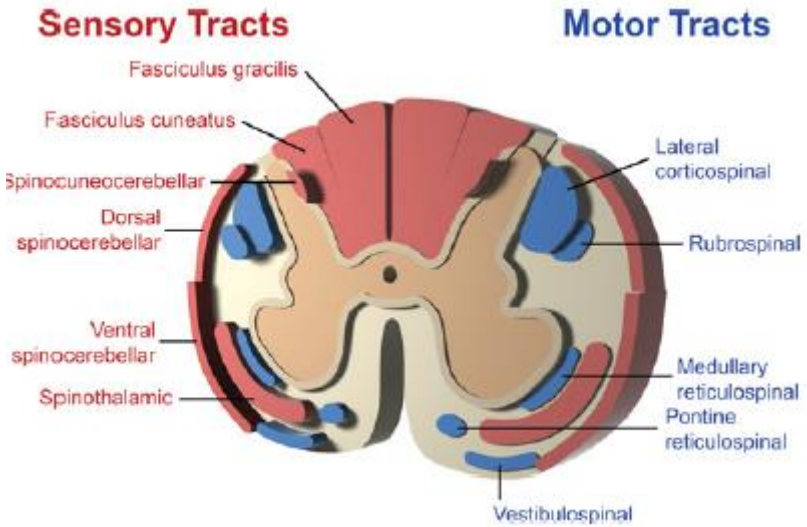




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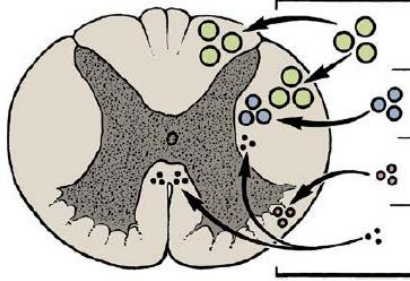
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Practical Guide to Canine and Feline Neurology, 3rd Edition

Handbook of Veterinary Neurology, 5th Edition



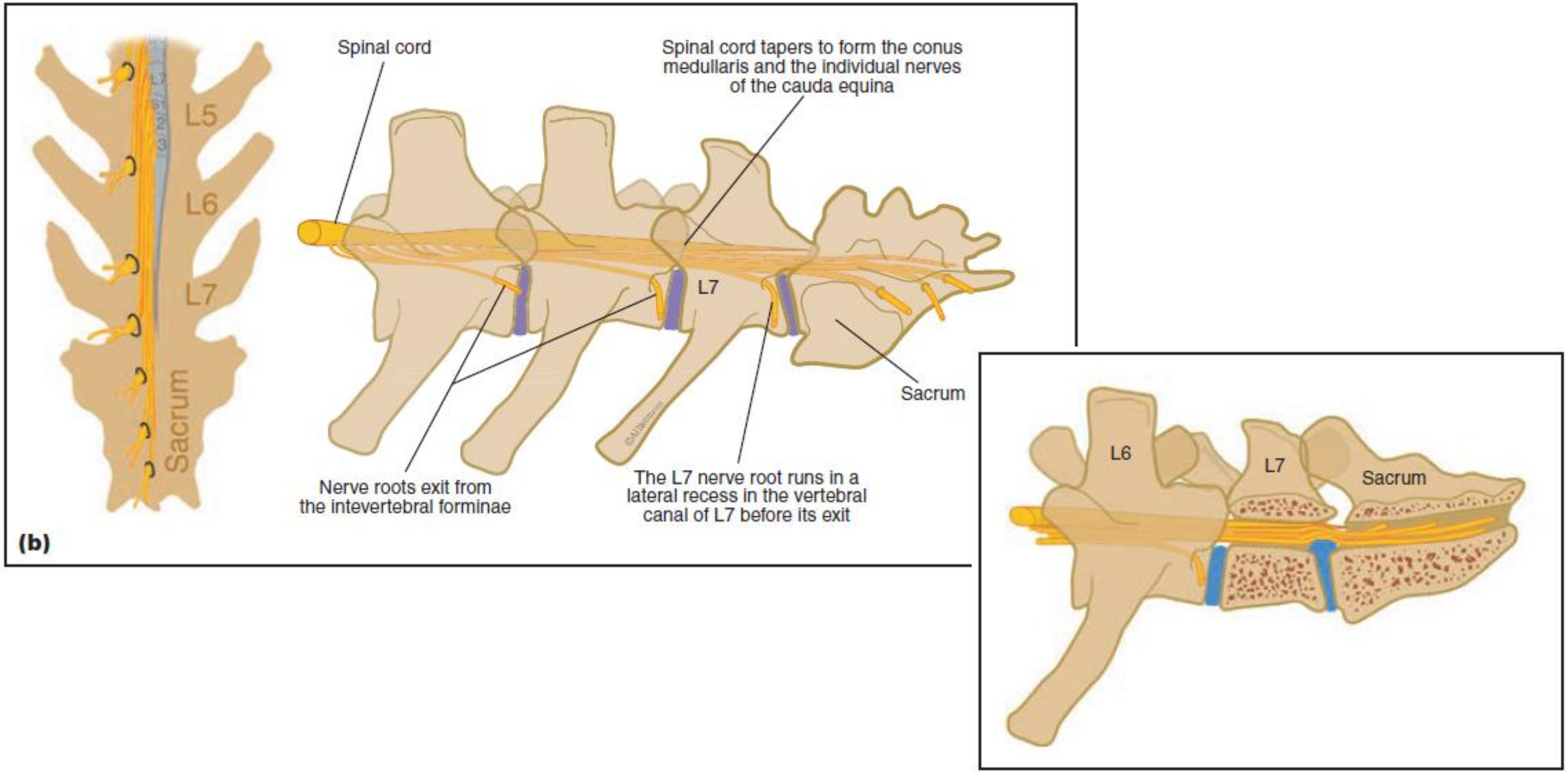
Fiber Size	Function	Signs with Increasing Compression	Prognosis
	General Proprioception	Proprioceptive Deficits	Good
	Voluntary Motor	Paresis, Paralysis	Fair
	Superficial Pain	Loss of Cutaneous Sensation	Fair
	Deep Pain	Loss of Deep Pain	Poor

Figure 1-3 Progression of signs in spinal cord compression.

https://www.safarivet.com/veterinary-services/intervertebral-disc-disease-anatomy-pathophysiology/

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病灶定位

Lesion Localization			
Lesion location 病灶區	Thoracic limbs 前肢	Pelvic limbs 後肢	Gaits 步態
Intracranial 顱內	Normal	Normal	Disorientation 東倒西歪
C1-C5	UMN上位神經元	UMN	Ataxia, long stride 大步
C6-T2	LMN下位神經元	UMN	Ataxia, two-engine 雙引擎
T3-L3	Normal	UMN	Ataxia, long stride
L4-S1	Normal	LMN	Ataxia, short stride 小步
S1-S3	Normal	LMN	Normal

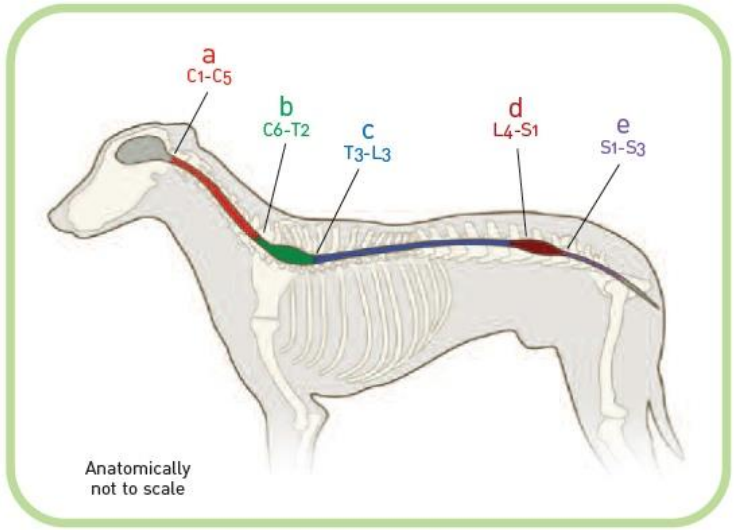
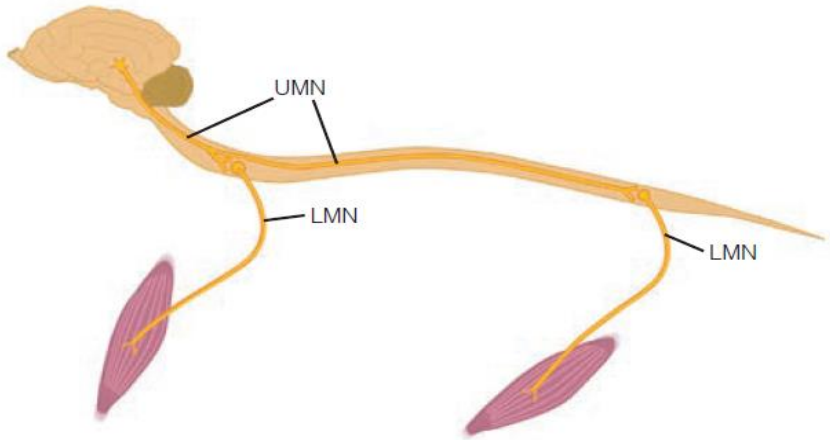
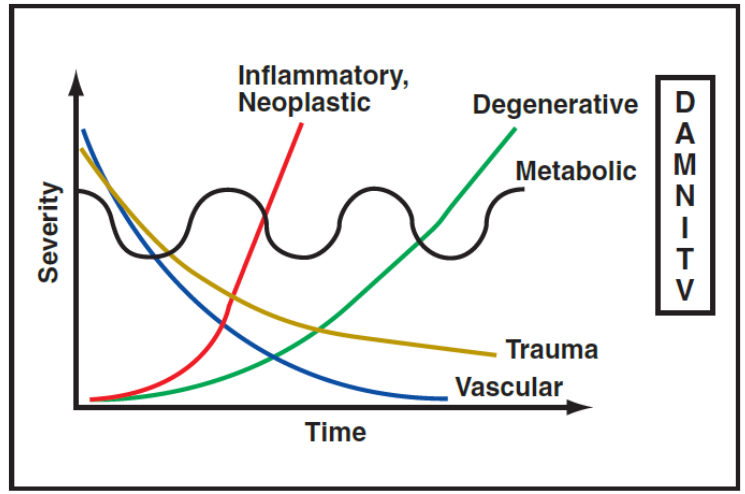


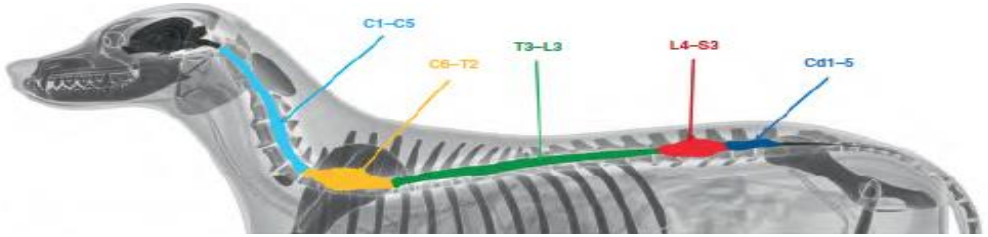
Fig 5. The functional divisions of the spine



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Cord Segments	C1–C5	C6–T2	T3–L3	L4–S3	Cd1–Cd5
Front Limb	**UMN** N to ↑ reflexes N to ↑ muscle tone Slow muscle atrophy Postural deficits often	**LMN** ↓ to absent reflexes ↓ to absent muscle tone Fast muscle atrophy Postural deficits often	**Normal**	**Normal**	**Normal**
Hind Limb	**UMN** N to ↑ reflexes N to ↑ muscle tone Slow muscle atrophy Postural deficits often	**UMN** N to ↑ reflexes N to ↑ muscle tone Slow muscle atrophy Postural deficits often	**UMN** N to ↑ reflexes N to ↑ muscle tone Slow muscle atrophy Postural deficits	**LMN** ↓ to absent reflexes ↓ to absent muscle tone Fast muscle atrophy Postural deficits	**Normal**
Bladder and Urination	**UMN** N to ↑ sphincter and detrusor tone Large, firm, difficult to express bladder Overflow incontinence Possible loss of voluntary voiding	**UMN** N to ↑ sphincter and detrusor tone Large, firm, difficult to express bladder Overflow incontinence Possible loss of voluntary voiding	**UMN** N to ↑ sphincter and detrusor tone Large, firm, difficult to express bladder Overflow incontinence Possible loss of voluntary voiding	**UMN: L4–S1(S2) OR LMN: S1–S3** Possible ↓ to absent sphincter and detrusor tone Possible large, flaccid bladder Possible constant leakage of urine due to overflow or pressure Possible absent voluntary voiding	**Normal**
Perianal Reflex and Defecation	**UMN** N perianal reflex and tone Likely voluntary defecation, Stool in rectum may stimulate reflex defecation	**UMN** N perianal reflex and tone Likely voluntary defecation Stool in rectum may stimulate reflex defecation	**UMN** N perianal reflex and tone Likely voluntary defecation Stool in rectum may stimulate reflex defecation	**LMN** Possible ↓ to absent perianal reflex and tone Possible absent voluntary defecation, continuous fecal leakage	**Normal**
Tail Tone and Movement	**UMN** N to ↑ tail tone and movement	**UMN** N to ↑ tail tone and movement	**UMN** N to ↑ tail tone and movement	**UMN** N to ↑ tail tone and movement	**LMN** Weak to absent tail tone and movement

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- **Neurodevelopmental sequence (NDS)**
  - The NDS is the logical steps of development that all dogs must go through to ambulate appropriately.
  - Patients must be able to transition from lateral to sphinx, sphinx to sit, sit to stand, and stand to walk.
  - 神經功能恢復順序
  - 由側躺➡ 趴正➡ 坐立➡ 站立➡ 行走

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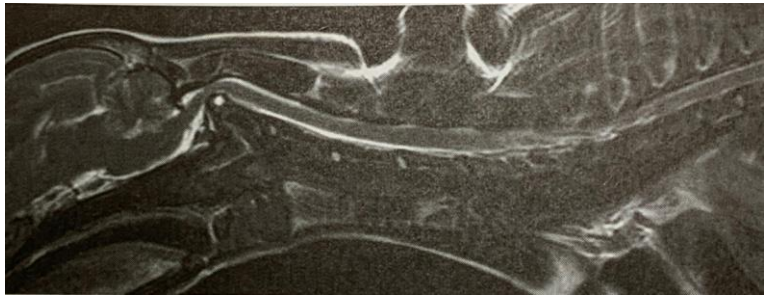
常見神經科疾病復健以及復健方案

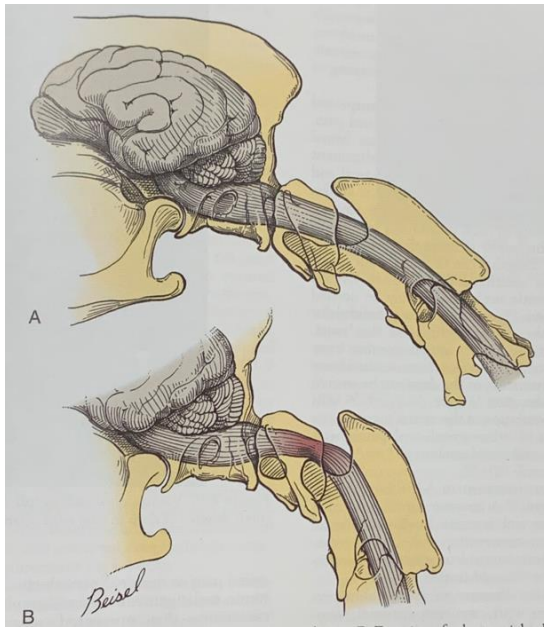
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- **A/A (C1/C2) instabilities 寰樞椎不穩**
  - Congenital
  - Traumatic
- Diagnosis
  - Clinical exam, radiograph, CT, MRI
- Treatment options
  - Surgery of fixation, External coaptation
  - Neck brace
  - Acupuncture, Herbal formula, Laser, PEMF and rehabilitation
- Rehabilitation plan
  - Acute phase (1st-5th day)- promote regeneration
    - Pain management, positioning, Massage, PROM, CP exercises
  - Proliferative phase (5th- 21st day)- exercises all the joints and limbs
    - Massage, PROM, weight shift, assisted standing
  - Consolidation phase (21st- 60th day) - recovering coordination and movements
    - Massage, PROM, wobble cushion, wobble board, assisted standing
  - Organization phase (61st ~day)- strength core and back muscles
    - No game , no stairs, AROM, sit to stand, down to sit, give-me-5,
    - Sling walk, UWTM elbow level

Problems
- ■ Acute
- ☐ Chronic
- ■ Pain
- ■ Weakness
- ■ CPD
- ■ Prognosis







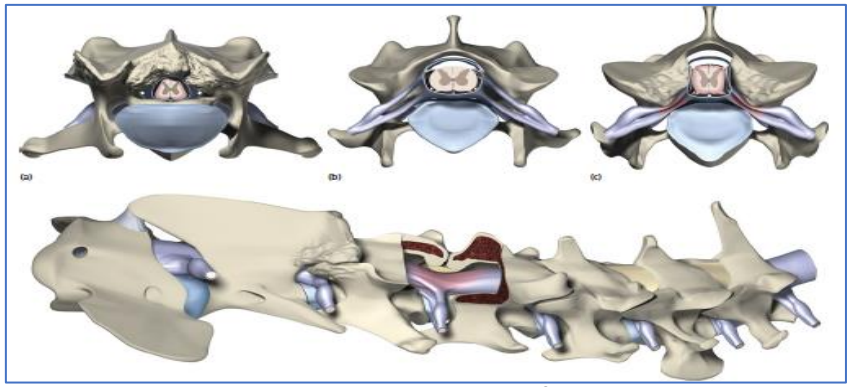


- **CCSM (Caudal cervical spondylomyelopathy)-wobbler syndrome 後頸椎脊髓病變–搖擺症**
  - Type 2 disc + Vertebral joint capsule + Ligament
  - Congenital- young dogs (Osseous-associated CSM, e.g. Great Danes, C1-C5)
  - Herniated disc- middle aged dogs (Disc-associated CSM, e.g. Doberman Pinscher, C6-T2)
- Diagnosis
  - Clinical exam, MRI is better than CT
- Treatment options
  - Surgery
  - Acupuncture, Herbal formula, Laser, PEMF and rehabilitation
- Rehabilitation plan
  - Acute phase (1st-5th day)- promote regeneration
    - Pain management, post- Sx swelling, Massage, PROM, CP exercises, positioning
  - Proliferative phase (5th- 21st day)- exercises all the joints and limbs
    - Massage, PROM, assisted standing, sit to stand with harness/sling, slow cookie stretch on neck
  - Consolidation phase (21st- 60th day) - recovering coordination and movements
    - AROM, rebounding, sling walk
  - Organization phase (61st ~day)- strength core and back muscles
    - UWTM

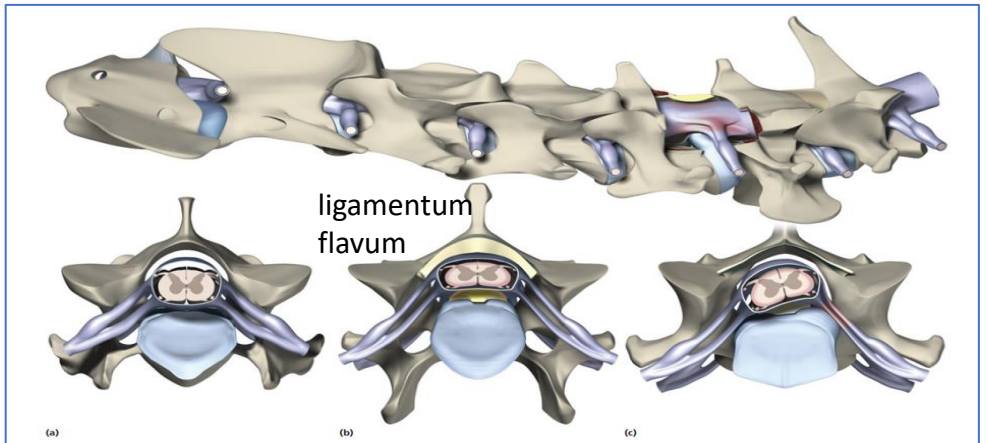
Problems
- ■ Acute
- ☐ Chronic
- ■ Pain
- ■ Weakness
- ■ CPD
- ☐ Prognosis



Osseous-associated CSM



Disc-associated CSM

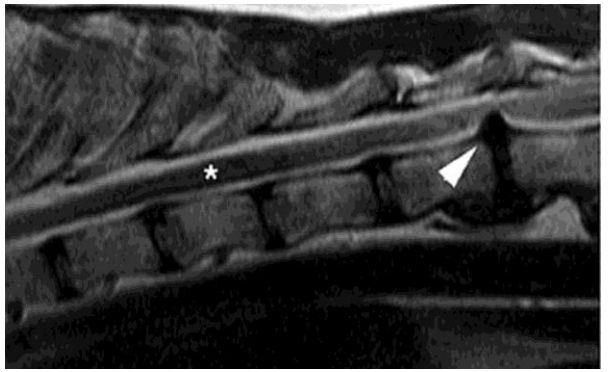


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- **IVDD type I (Hanson type 1; extrusion) 椎間盤破裂**
  - Chondrodystrophic breeds e.g. Dachshund
  - Paraparesis, tetraparesis, -plegia, -paralysis
  - Surgical case
  - Deep pain loss (Prolonged loss of nociception > 48 hours) the prognosis poor
- Diagnosis
  - Clinical exam, radiograph, CT, MRI
- Treatment options post Sx
  - Acupuncture, Herbal formula, Laser, and rehabilitation
- Rehabilitation plan
  - Acute phase (1st-5th day)- promote regeneration
    - Pain management, wound swelling care, Massage, PROM, stimulate reflex of paws, weight shift, stand
  - Proliferative phase (5th- 21st day)- exercises all the joints and limbs
    - Massage, PROM, slow walking, UWTM (if motor movement present)
  - Consolidation phase (21st- 60th day) - recovering coordination and movements
    - Strengthening muscles, short intervals of UWTM, weave poles
  - Organization phase (61st ~day)- strength core and back muscles
    - Increase the intensity of exercises, Swimming

Problems
- ■ Acute
- ☐ Chronic
- ■ Pain
- ■ Weakness
- ■ CPD
- ■ Prognosis





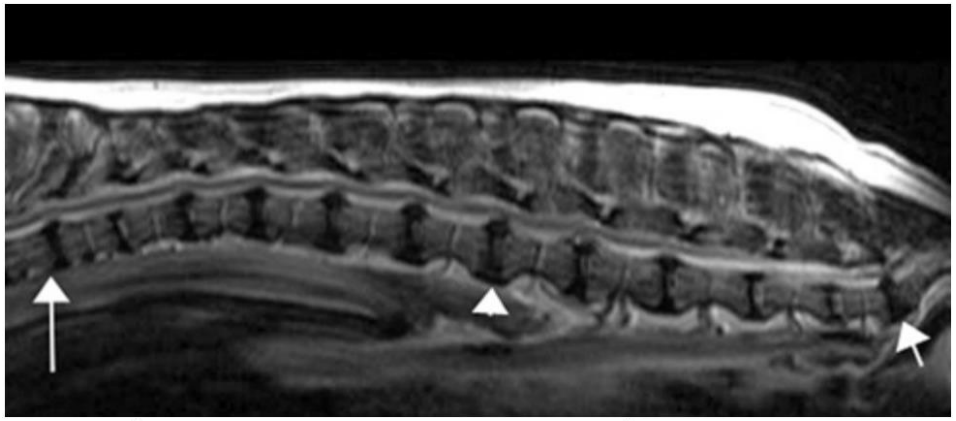




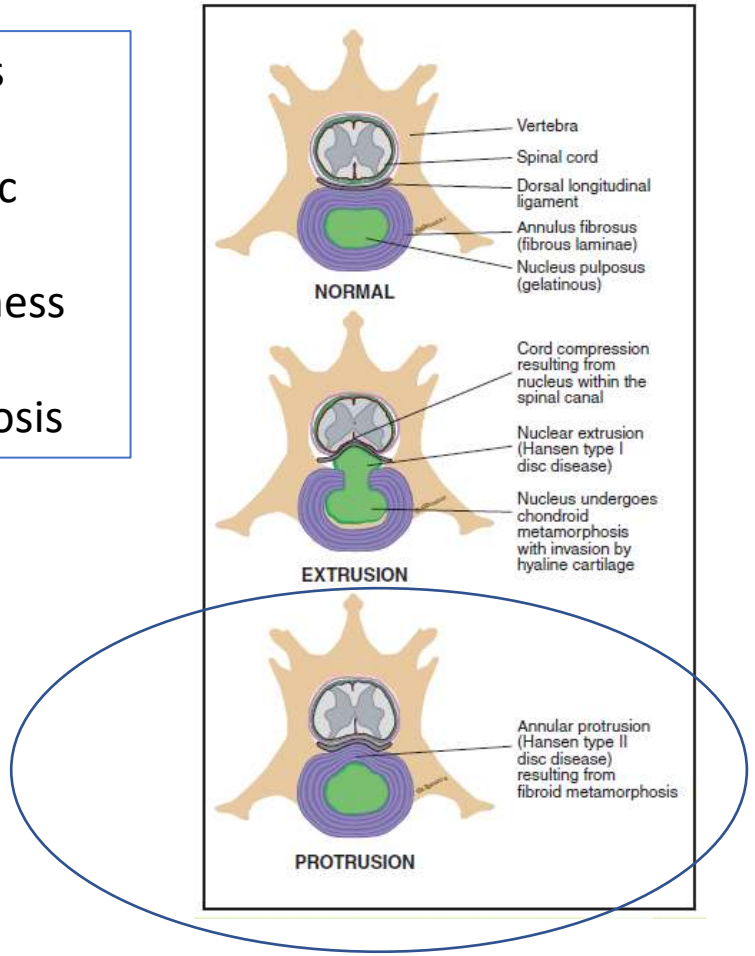


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- **IVDD type II (Hanson type II; protrusion)椎間盤突出**
  - Non-chondrodystrophic breeds
  - Thoracolumbar
  - 10% have back pain with no neurological deficits
  - Non-surgical treatments
- Diagnosis
  - Clinical exam, radiograph, CT, MRI
- Treatment options
  - Acupuncture, Herbal formula, Laser, and rehabilitation
- Rehabilitation plan
  - Acute phase (1st-5th day)- promote regeneration
    - Pain management, Massage, PROM, CP exercises, avoid motions
  - Proliferative phase (5th- 21st day)- exercises all the joints and limbs
    - Massage, PROM, CP exercises
  - Consolidation phase (21st- 60th day) - recovering coordination and movements
    - Treadmill, UWTM, Cavaletti rails
  - Organization phase (61st ~day)- strength core and back muscles
    - Swimming, UWTM, life-long therapy



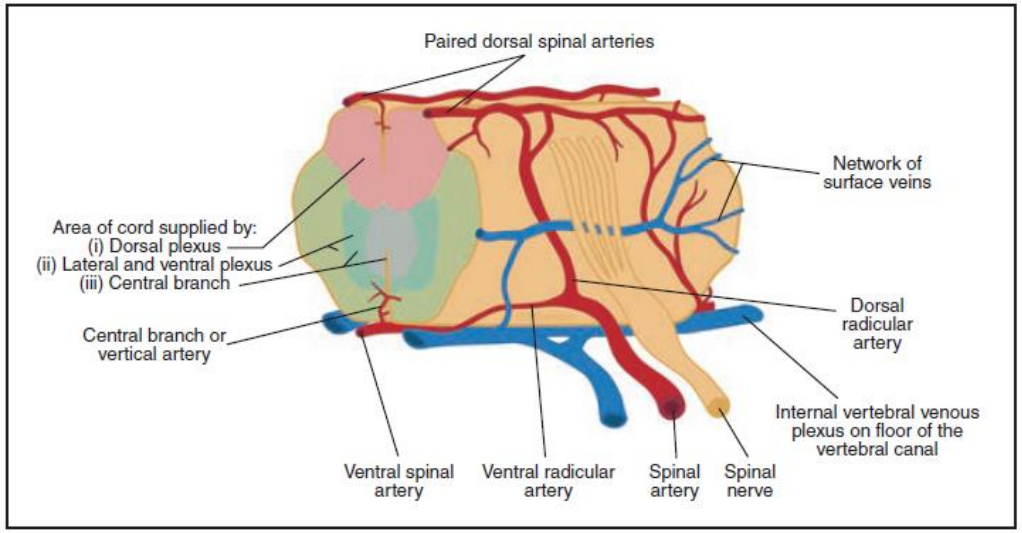
Problems
- ☐ Acute
- ■ Chronic
- ■ Pain
- ■ Weakness
- ■ CPD
- ☐ Prognosis

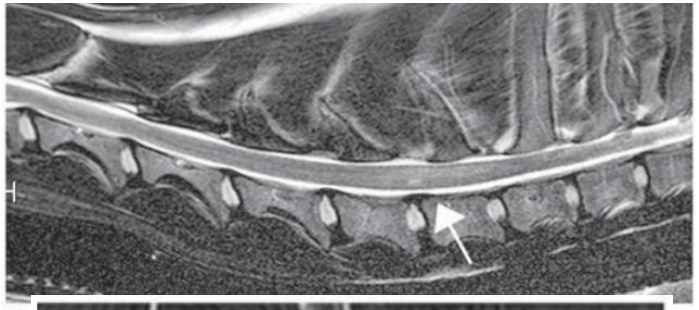


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- **FCE (fibrocartilaginous embolism) 纖維軟骨性栓塞**
  - Embolization of the arterial and/or venous blood supply of a section of the spinal cord
  - Paraparesis, tetraparesis, -plegia
  - The animal often yelp at the moment of embolus occurs
  - Not a surgical case
- Diagnosis
  - Clinical exam, MRI
- Treatment options
  - Acupuncture, Herbal formula, Laser, and rehabilitation
- Rehabilitation plan
  - Acute phase (1st-5th day)- promote regeneration
    - Massage, PROM, CP exercises
  - Proliferative phase (5th- 21st day)- exercises all the joints and limbs
    - Massage, PROM, weight shift, UWTM (if motor movement)
  - Consolidation phase (21st- 60th day) - recovering coordination and movements
    - Treadmill, cavaletti rails,
  - Organization phase (61st ~day)- strengthen core and back muscles
    - Swimming, UWTM

Problems
- ■ Acute
- ☐ Chronic
- ☐ Pain
- ■ Weakness
- ■ CPD
- ☐ Prognosis







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Compressive

- **Hansen type I (Extrusion)椎間盤破裂**
- **Hansen type II (Protrusion)椎間盤突出**
- **Hydrated nucleus pulposus extrusion (HNPE)水合髓核脫出**
  - Hydrated extradural disc material compressing spinal cord

Non-compressive

- **Acute Non-compressive Nucleus Pulposus Extrusion (ANNPE)急性非壓迫性髓核脫出**
  - Traumatic spinal cord contusion
  - Also known as “high-velocity-low- impact disc extrusion” or “**Hansen type III** disc disease”
- **Fibrocartilaginous Embolism (FCE)纖維軟骨性栓塞**
  - Also referred to as “ischemic myelopathy) ”

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STANDARD ARTICLE OPEN ACCESS

Small Animal Internal Medicine Respiratory

Exercise Restriction Does Not Change Outcome in Dogs After Diagnosis of Acute Non-Compressive Nucleus Pulposus Extrusion, Fibrocartilaginous Embolism, or Hydrated Nucleus Pulposus Extrusion

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2025

ABSTRACT

Background: Acute non-compressive nucleus pulposus extrusion (ANNPE), fibrocartilaginous embolism (FCE), and hydrated nucleus pulposus extrusion (HNPE) all present with acute to peracute onset of myelopathic signs in dogs, for which treatment usually is medical. Conservative management including a period of strict rest usually is recommended because of concern about relapse if immediate exercise is allowed.

Hypotheses:

1. Allowing exercise after a diagnosis of ANNPE, FCE, or HNPE does not decrease the chance of recovery or predispose to relapse in the short-term follow-up period (4 weeks) after discharge.
2. There is no difference in clinical outcome between rested and exercised groups.

Animals: Forty cases of ANNPE, FCE, or HNPE with follow-up including at minimum a 4-week evaluation after discharge.

Methods: Retrospective study. The exercise group ($n = 22$) included all dogs the owners of which were not explicitly instructed to rest their dogs, whereas the rest group ($n = 18$) included all dogs the owners of which were specifically advised to rest their dogs or to restrict their exercise.

Results: No dogs relapsed or deteriorated between discharge and 4-week re-evaluation. No significant difference was found between the rest and exercise groups.

Conclusions: Allowing exercise after confirming a diagnosis of ANNPE, FCE, or HNPE did not predispose to relapse of clinical signs in the 4 weeks after discharge in our cohort, but the rest group had a higher proportion of HNPE cases, which means interpretation of our findings in this diagnosis must be made with caution.

- **ANNPE** (Acute Non-Compressive Nucleus Pulposus Extrusion)
  急性非壓迫性髓核脫出
- **HNPE** (Hydrated Nucleus Pulposus Extrusion)
  水合髓核脫出
- **FCE** (Fibrocartilagous Embolism)
  纖維軟骨性栓塞

- 病例回溯性研究針對40隻家犬罹患ANNPE or HNPE or FCE出院後4週內疾病復發以及復原情形
- 一組為嚴格限制運動及休息(n=18)與另一組沒特別限制活動(n=22)，兩組做比較其復發以及復原程度並無差異
- 結論:
- 讓上述動物運動並不影響其預後

- **Degenerative Myelopathy 退化性脊髓病**
  - Hereditary, breed predisposition
  - T3-L3 area
  - Progressive degeneration
- Diagnosis
  - Clinical exam, MRI and CT are usually unremarkable, but often other concomitant diseases, such as herniated discs, are diagnosed
  - Gene test
  - Exclusive diagnosis
- Treatment options
  - Not a surgical case
  - Median survival time 6-12 months
  - Acupuncture, Herbal formula, Laser, and rehabilitation
- Rehabilitation plan
  - Massage, PROM, CP exercises, UWTM, swimming

Problems
☐ Acute
■ Chronic
☐ Pain
■ Weakness
■ CPD
■ Prognosis



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- **Brachial Plexus avulsions 臂神經叢撕裂**
  - Brachial plexus originate from the spinal segments C6-T2
  - Strong backwards motion with simultaneous abduction of the forelimb e.g. hit by car
  - Complete (C6-T1) or partial (C6-C7, or C8-T1) avulsions
- Diagnosis
  - Clinical exam, EMG, CT, MRI
- Treatment options
  - Not a surgical case
  - Amputation
  - Rehabilitation

Problems
- ■ Acute
- ☐ Chronic
- ☐ Pain
- ■ Weakness
- ■ CPD
- ■ Prognosis

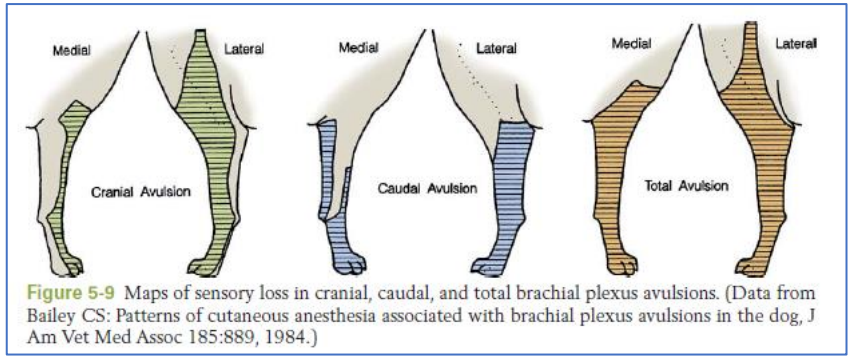
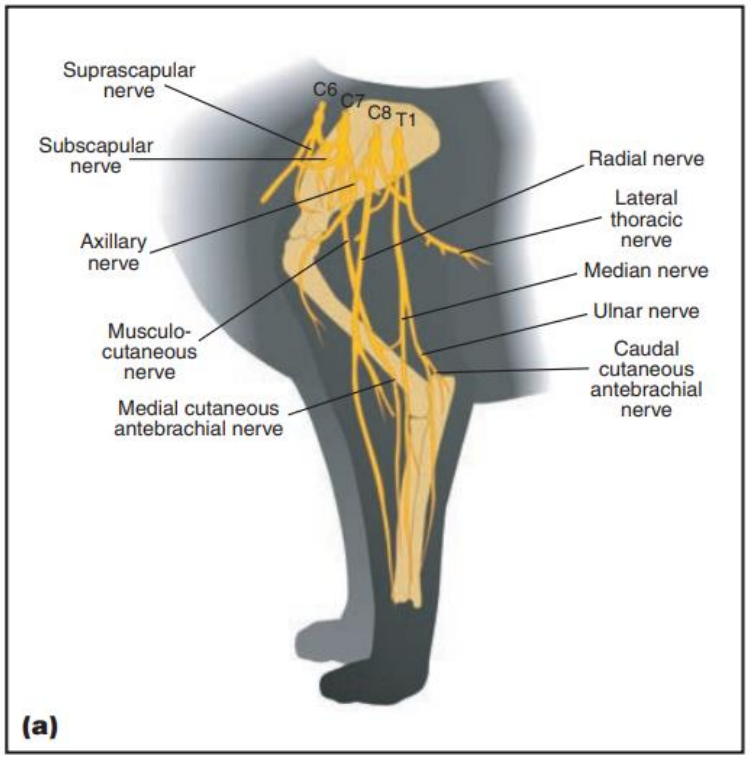


Figure 5-9 Maps of sensory loss in cranial, caudal, and total brachial plexus avulsions. (Data from Bailey CS: Patterns of cutaneous anesthesia associated with brachial plexus avulsions in the dog, J Am Vet Med Assoc 185:889, 1984.)



- Rehabilitation plan
  - Acute phase (1st-5th day)- promote regeneration
    - Massage, PROM
  - Proliferative phase (5th- 21st day)- exercises all the joints and limbs
    - Massage, PROM
  - Consolidation phase (21st- 60th day) - recovering coordination and movements
    - Massage, PROM, stretching
  - Organization phase (61st ~day)- strength core and back muscles
    - Pawing at a treat dangling in front of them, give me five exercises, climbing upstairs

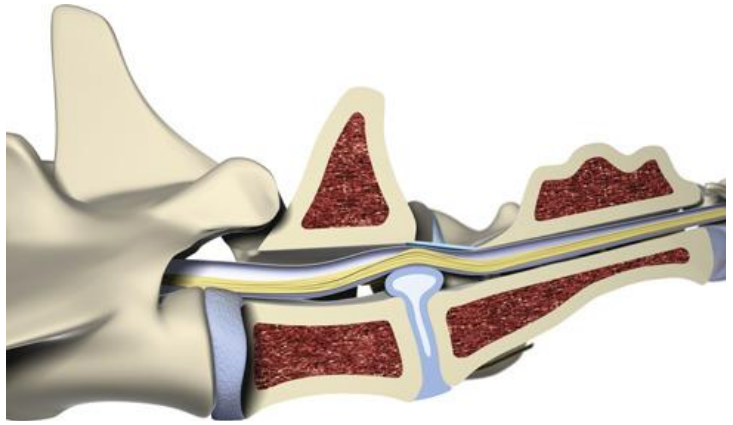




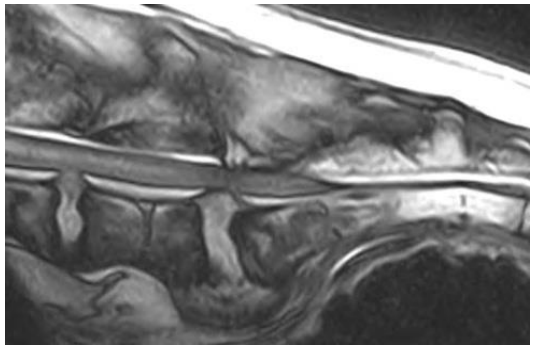
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- **Caudal Equina Syndrome 馬尾部症候群**
  - Nerve root compression
  - Degenerative L/S stenosis (DLSS 退行性腰薦椎狹窄)
  - Incontinence
  - Tail carriage low
- Diagnosis
  - Clinical exam, CT, MRI
- Treatment options
  - Surgery
  - Acupuncture, Herbal formula, Laser, PEMF and rehabilitation
- Rehabilitation plan
  - Acute phase (1st-5th day)- promote regeneration
    - Pain management, post- Sx swelling, Massage, PROM, CP exercises
  - Proliferative phase (5th- 21st day)- exercises all the joints and limbs
    - Pain management, post- Sx swelling, Massage, PROM, CP exercises
  - Consolidation phase (21st- 60th day) - recovering coordination and movements
    - AROM, treadmill, UWTM, cavaletti rails
  - Organization phase (61st ~day)- strength core and back muscles
    - UWTM, swimming

Problems
- ☐ Acute
- ■ Chronic
- ■ Pain
- ■ Weakness
- ■ Lameness
- ■ CPD
- ☐ Prognosis





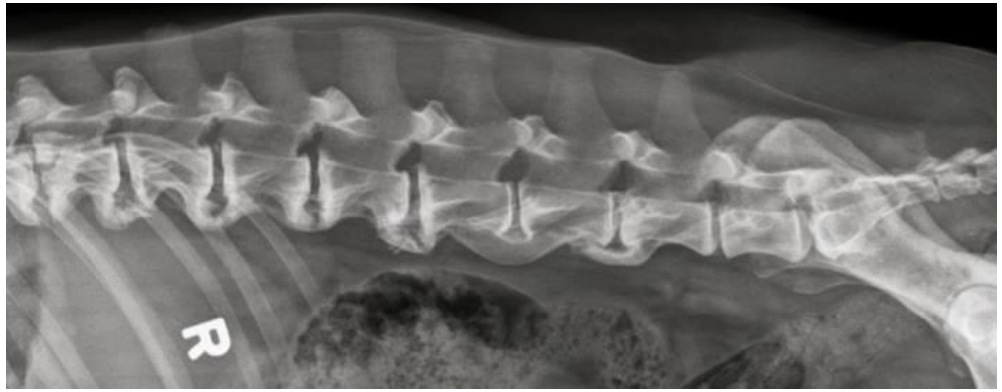




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- **Spondylosis deformans 變形性脊椎病;骨刺**
  - Formation of osteophytes on vertebral bodies
  - Compression of spinal nerves
  - Painful muscle tensions
- Diagnosis
  - Clinical exam, radiographs
- Treatment options
  - Not a surgical case
  - Acupuncture, Herbal formula, Laser, and rehabilitation
- Rehabilitation plan
  - Acute phase (1st-5th day)- promote regeneration
    - Massage, PROM, simple CP exercises
  - Proliferative phase (5th- 21st day)- exercises all the joints and limbs
    - Massage, PROM, short intervals of UWTM and control walk
  - Consolidation phase (21st- 60th day) - recovering coordination and movements
    - UWTM, walking on a variety of surfaces
  - Organization phase (61st ~day)- strength core and back muscles
    - Swimming, UWTM, walk

Problems
☐ Acute
■ Chronic
■ Pain
■ Weakness
☐ CP
■ Prognosis







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- **Acute idiopathic polyradiculoneuritis (coonhound paralysis)**
- **急性特發性多發性神經根炎（浣熊犬麻痺症）**
  - Idiopathic inflammation of axons and myelin of the ventral nerve roots
  - Ascending para- or tetra- paresis or plegia with LMN sign
  - Spontaneous healing within 3-6 weeks
- Diagnosis
  - Clinical exam, cerebral spinal fluid (CSF), electrodiagnostics
- Treatment options
  - Not a surgical case
  - Nursing care
  - Acupuncture and rehabilitation
- Rehabilitation plan
  - Acute phase (1st-5th day)- elicit the flexor reflex
    - Pain management, Massage, PROM, CP stimulation exercises
  - Proliferative phase (5th- 21st day)- positioning and standing
    - Standing, walking exercises in water,
  - Consolidation phase (21st- 60th day) - recovering coordination and movements
    - UWTM, land treadmill, exercise with stability ball, NMES
  - Organization phase (61st ~day)- strength core and back muscles
    - Swimming, UWTM, walk

Problems
- ■ Acute
- ☐ Chronic
- ■ Pain
- ■ Weakness
- ■ CP
- ☐ Prognosis

TABLE 7-8

Diagnostic Comparison of Acute Progressive Lower Motor Neuron Disorders

	Polyradiculoneuritis	Tick Paralysis	Botulism
History	Single case; previous exposure to raccoon in some cases	Single case—engorged tick	Multiple cases very suggestive; access to carrion or spoiled food
Pathophysiology	Nonsuppurative nerve root inflammation and demyelination	Interference with action potential or blocks release of acetylcholine	Toxin blocks release of acetyl-choline by altering docking of synaptic vesicles
Esophageal motility	Normal	Normal	Reduced to absent
Hyperesthesia	Frequently present	Absent	Absent
Anal sphincter tone	Normal	Normal	Reduced to absent
Autonomic signs	Rare	Rare	Common
Cranial nerve involvemen	Rare (sometimes CN VII)	Rare	Usually (often involving tongue and (pharynx)
Electromyography	Fibrillation potentials and positive sharp waves	No denervation	Usually no denervation
Conduction velocity	Normal to decreased	Normal to slightly decreased	Normal
Evoked potentials	Reduced	Reduced	Reduced; decrement of amplitude on repetitive stimulation
Special tests	None	None	Toxin in feces and serum
Treatment	Supportive	Tick removal	Supportive
Recovery time	3-6 wk	24-48 hr	2-3 wk

Handbook of Veterinary Neurology, 5th Edition

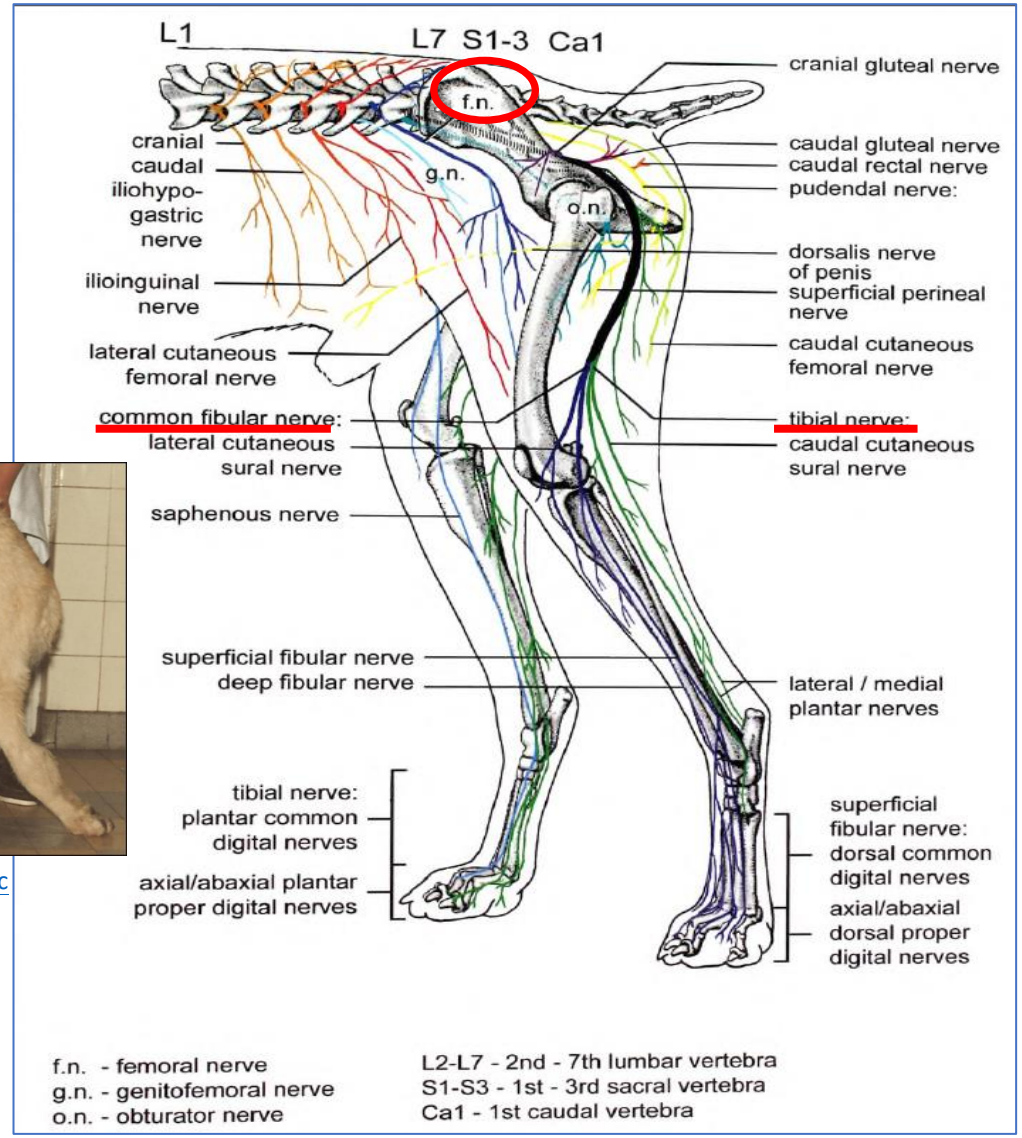
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- **Sciatic nerve injury 坐骨神經損傷**
  - The sciatic nerve is composed of a collection of nerves arising from the spinal cord segments of L6-S2.
  - The extensors of the stifle (innervated by the femoral nerve) are the only muscles that remain functional.
  - The animal may bear weight on the limb, but they can't flex the stifle nor can they flex or extend the tarsus and digits.
  - Thus, the animal has knuckling of the digits and a plantigrade stance.
  - Severe muscle atrophy
  - If the nerve is impinged, severe pain results.
- Diagnosis
  - Clinical exam, EMG
- Treatment options
  - Either Sx or conservative Tx
  - Acupuncture, Laser, and rehabilitation
- Rehabilitation plan
  - Wear booties, orthotics, Massage, PROM, CP and pattern exercises, UWTM
  - Dogs that only have a tibial nerve injury function better than those that have sciatic or peroneal nerve involvement. Many injuries are caused by neurapraxia or axonotmesis and the animal regains functional use of the limb.

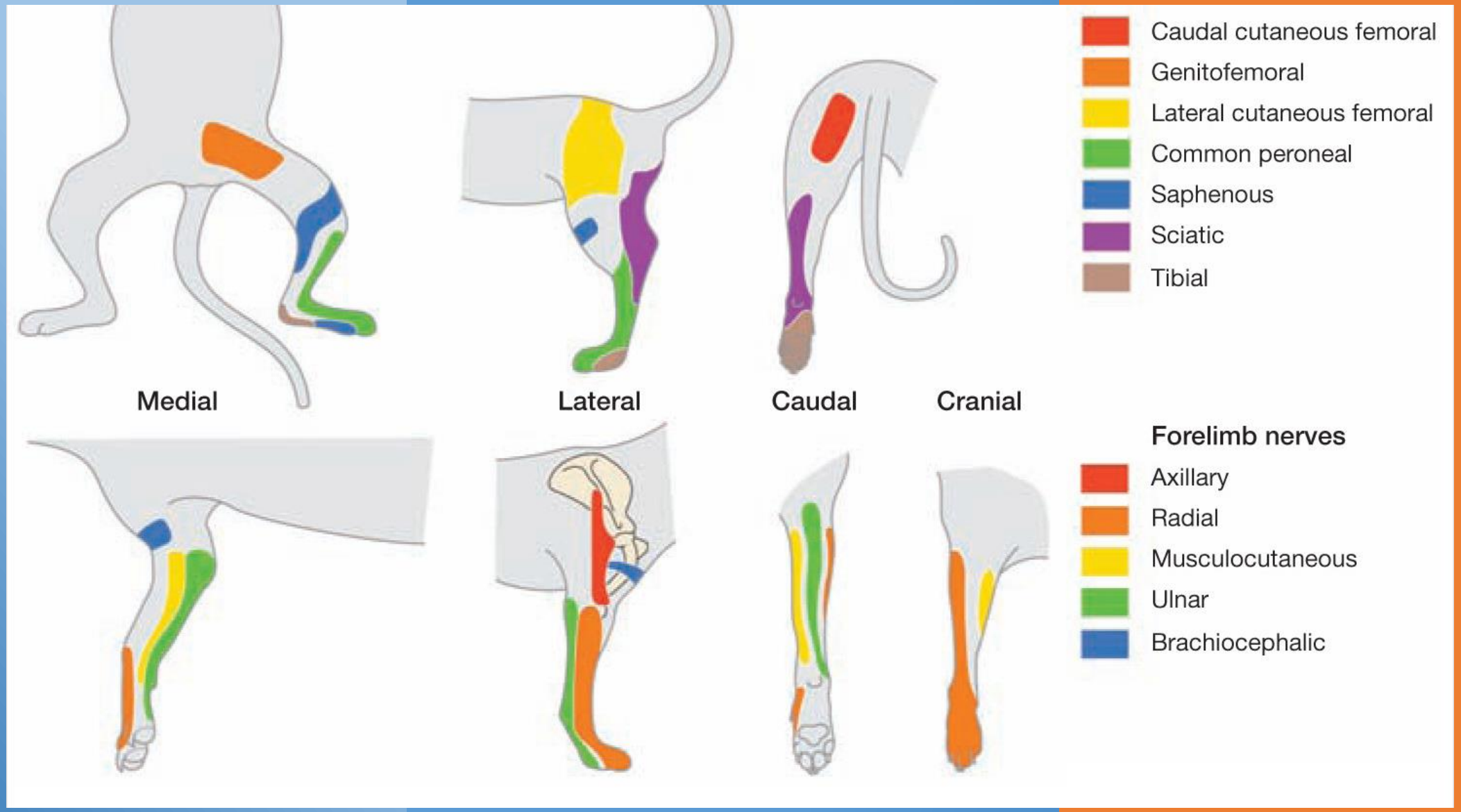
Problems
- ■ Acute
- □ Chronic
- ■ Pain
- ■ Weakness
- ■ CP
- ■ Prognosis



https://www.researchgate.net/publication/292349439



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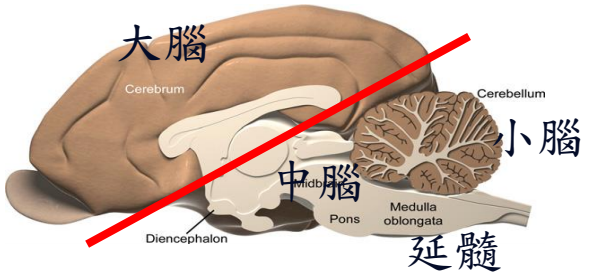


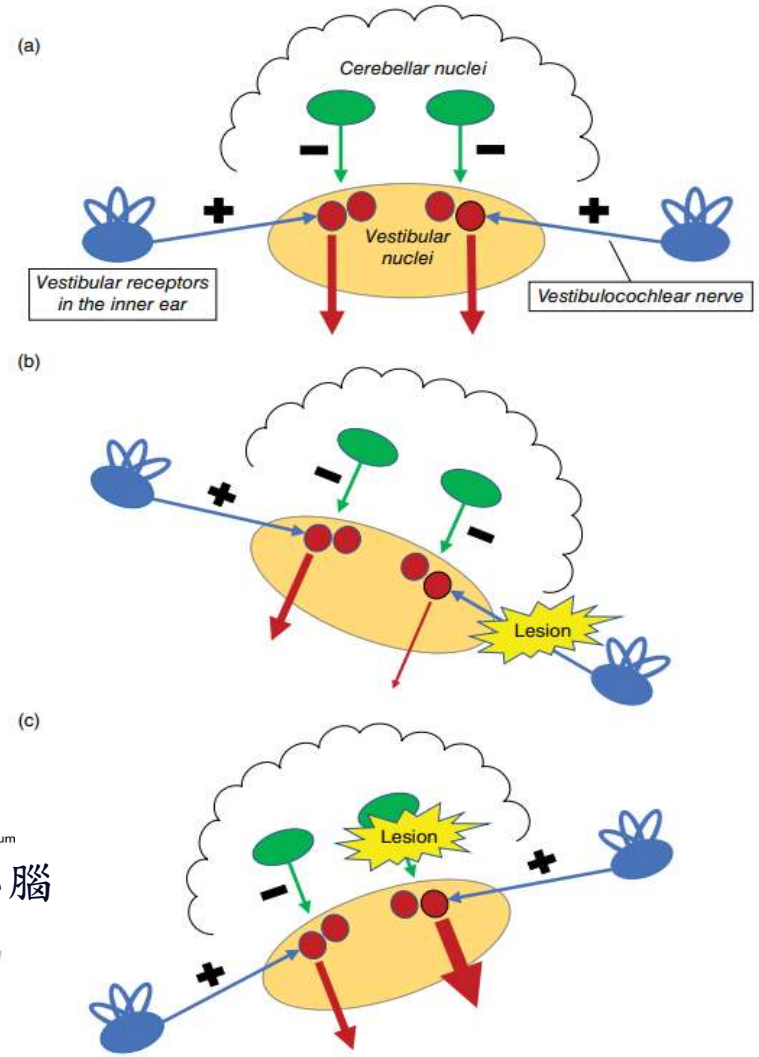
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- **Vestibular disease**
  - An idiopathic disease occurring in mostly large breed dogs more than eight years old.
  - The owner often falsely describes the condition as a "stroke."
  - Severe ataxia to the point that the dog falls over
  - Head tilt, Nystagmus without Horner's Syndrome, Vomiting
  - Paradoxical
- Diagnosis
  - Clinical exam, exclusion of other disorders
- Treatment options
  - Palliative, antiemetics
  - Acupuncture and rehabilitation
- Rehabilitation plan
  - Acute phase (1st-5th day)- elicit the flexor reflex
    - Massage, PROM, rest and prevent getting excited
  - Proliferative phase (5th- 21st day)- positioning and standing
    - Stimulate flexor reflex, standing exercises
  - Consolidation phase (21st- 60th day) - recovering coordination and movements
    - Slow walk and walk in UWTM coordinated movement and CP exercises
  - Organization phase (61st ~day)- strength core and back muscles
    - Increase the intensity of exercises, Unstable movement patterns

Problems
- ■ Acute
- ☐ Chronic
- ☐ Pain
- ☐ Weakness
- ■ CP
- ☐ Prognosis







Freeman, 2020, A Practical Approach to Neurology for the Small Animal Practitioner.

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甚麼是癱瘓(paralysis)?

- 一般所指的癱瘓是脊髓損傷或壓迫導致肢體自主動作功能發生問題，程度上分輕癱(paresis)、截癱(plegia)、癱瘓(paralysis)
- 有四肢同時發生(tetra-)、有半邊偏癱(hemi-) 、有後半身軀癱(para-)，常與發生疾病病灶部位有關
- 無深層痛覺表示預後不佳
- Ataxia
- weakness
- UMN
- LMN

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癱瘓診斷

- Neurological examination
- Localization
  - CNS (central nerve system)
    - Intracranial, C1-C5, C6-T2, T3-L3, L4-S1~3
  - PNS(peripheral nerve system)
  - NMJ(neuromuscular junction)
  - Muscle
- Differential diagnosis
- Blood test, CT, MRI, CSF analysis, special test
- Final diagnosis

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癱瘓復健治療與照顧

- 復健治療:
  - Acupuncture
  - Laser
  - tPEMF
  - HBOT
- 肢體復健方案
- 運動復健
- 輔具使用
- 居家環境調整

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針灸治療

Original Article

J Vet Sci 2016, 17(3), 337-345 · http://dx.doi.org/10.4142/jvs.2016.17.3.337

JVS

Retrospective study of the clinical effects of acupuncture on cervical neurological diseases in dogs

Ching Ming Liu[1], Fang Chia Chang[2], Chung Tien Lin[3,*]

[1]*National Taiwan University Veterinary Hospital, Taipei 10617, Taiwan*
[2]*Department and Graduate Institute of Veterinary Medicine, and* [3]*Institute of Veterinary Clinical Sciences, School of Veterinary Medicine, National Taiwan University, Taipei 10617, Taiwan*

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- 四肢癱瘓
- **Treatment option 1**:

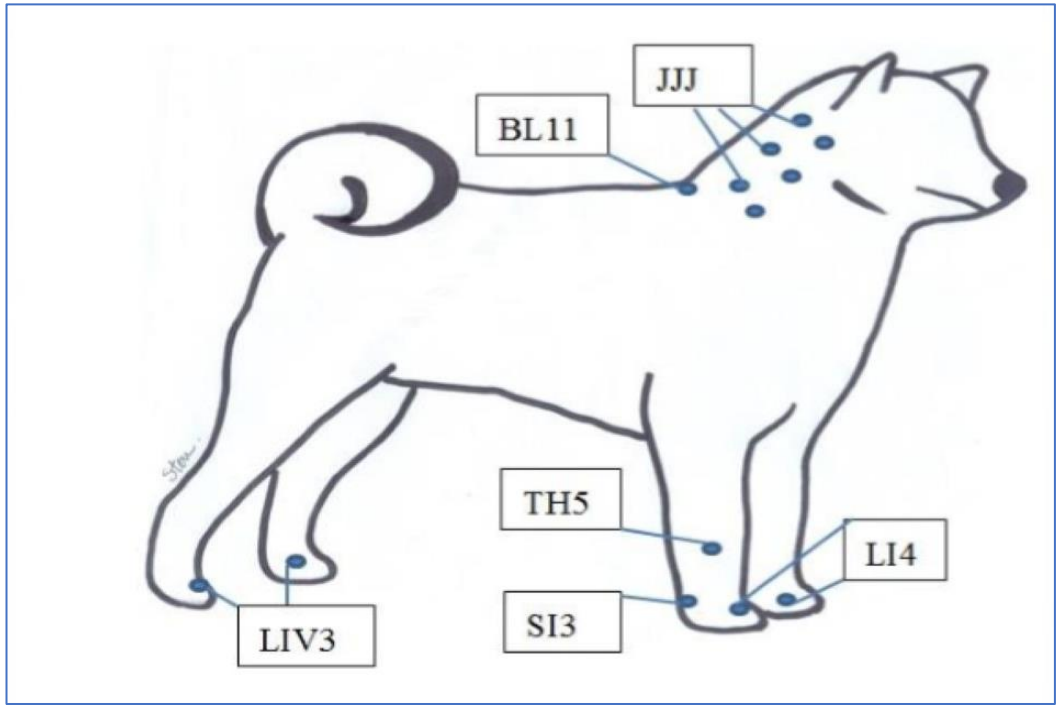
AP in local points C1-C3-C5 (JJJ) and in distant points SI-3, LIV-3, LI-4

- **Treatment option 2**:

Option 1 with EAP in local points JJJ.

- **Treatment option 3**:

Option 2 with EAP in distant points BL-11+TH-5 (BL11 to TH5 connected).



AP: place needles 15 minutes.

EAP: electroacupuncture 20Hz for 15 minutes

Twice a week

Herbal medicine:

℞ Double PII, Cervical Formula

1 gm/5-10 kg, bid, PO

Acupoint	Name (Chinese)	Name (English)
JJJ	頸夾脊穴	Jing-jia-ji (C1-C3-C5)
BL-11	大杼	Da-zhu
TH-5	外關	Wai-guan
LI-4	合谷	He-gu
LIV-3	太衝	Tai-chong
SI-3	後谿	Hou-xi

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Pakistan Veterinary Journal

ISSN: 0253-8318 (PRINT), 2074-7764 (ONLINE)
Accessible at: www.pvj.com.pk

RESEARCH ARTICLE

Retrospective Study of a New Standardized Acupuncture Treatment Protocol on Thoracolumbar Spinal Cord Diseases in 84 Dogs

Ching Ming Liu[1] and Chung Tien Lin[2]*

[1]National Taiwan University Veterinary Hospital, Da'an Dist., Taipei 106; [2]Institute of Veterinary Clinical Sciences, School of Veterinary Medicine, National Taiwan University, Taipei 106, Taiwan, R.O.C.
*Corresponding author: ctlin@ntu.edu.tw

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To Cite This Article: Liu CM and CT Lin, 2015. Retrospective study of a new standardized acupuncture treatment protocol on thoracolumbar spinal cord diseases in 84 dogs. Pak Vet J, 35(4): 461-465.

- **後軀癱瘓**
- **Treatment option 1**:

AP in local HTJJ points and distant points LIV-3, LI-4 for 15 minutes

- **Treatment option 2**:

Option 1 with the addition of distant points GB-30, GB-34

- **Treatment option 3**:

Option 2 with the addition of EAP in local HTJJ points, distant points GB-30 + GB-34 (connects GB-30 to GB-34)

- **Treatment option 4**: Option 3 with the addition of the hind limb *Liu-feng* points.

The *Liu-feng* acupoints were added to generate the strongest stimulation to the extremity of the limbs.

AP: place needles 15 minutes.

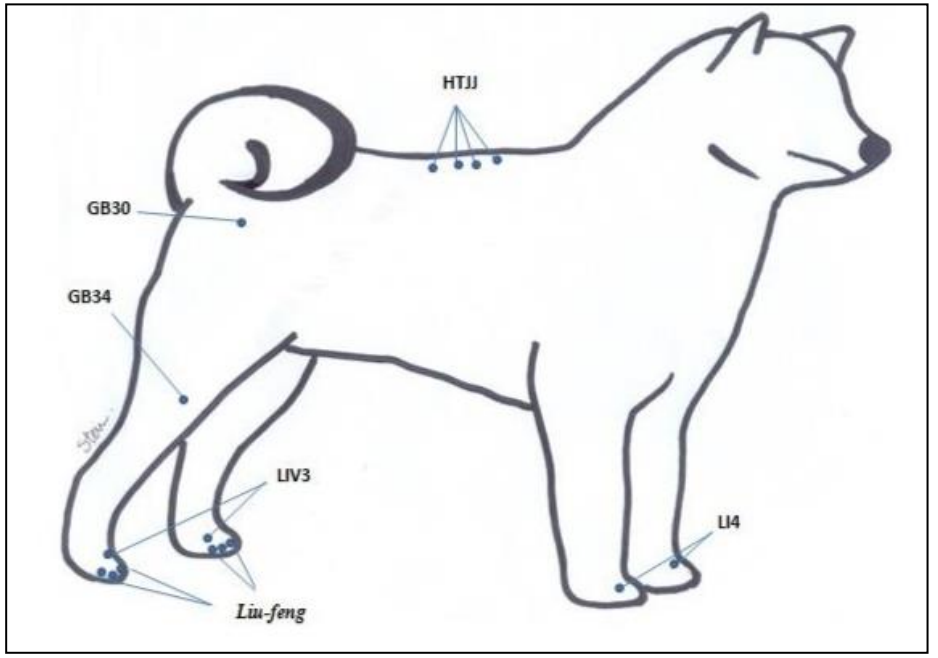
EAP: electroacupuncture 20Hz for 15 minutes

Twice a week

Herbal medicine:

℞ Double PII, Bu Yang Huan Wu

1 gm/5-10 kg, bid, PO



Acupoint	Name (Chinese)	Name (English)
HTJJ	華佗夾脊穴	Hua-tuo-jia-ji
GB-30	還跳	Huan-tiao
GB-34	陽陵泉	Yang-ling-quan
LI-4	合谷	He-gu
LIV-3	太衝	Tai-chong
Liu-feng	六縫	*Liu-feng*

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Laser雷射治療

Perilesional photobiomodulation therapy and physical rehabilitation in post-operative recovery of dogs surgically treated for thoracolumbar disk extrusion

Enrico Bruno[1], Sara Canal[1], Michela Antonucci[1], Marco Bernardini[1,2], Federica Balducci[1], Vincenzo Musella[3], Matteo Mussoni[4] and Giuseppe Spinella[5*]

2020

Abstract

Background: Recent studies have reported contrasting results of the effects of laser therapy on post-operative intervertebral disk herniation, with a lack of evidence-based advantages of this modality within a rehabilitation protocol.

The aim of this study was to report the clinical effects of photobiomodulation therapy within a post-operative rehabilitation protocol in dogs submitted to surgery for thoracolumbar disk extrusion. Twenty-four dogs were included in the study (12 dogs treated with laser therapy and rehabilitation protocol and 12 dogs treated with same rehabilitation protocol but without laser therapy).

Results: All dogs treated with laser therapy showed improved neurological status (Modified Frankel Score more than 3 within 30 days of physiotherapy starting) if deep nociception on admission was maintained ($P = 0.04$). However, Kaplan-Meier analysis did not show any statistical difference in time to regain ambulatory ability, although there was a tendency for a shorter mean time of 14.2 ± 8.55 days in the laser group versus 24 ± 18.49 days in the no laser group.

Conclusions: The use of laser therapy in the post-operative rehabilitation of dogs affected by intervertebral disc extrusion and submitted to surgery for spinal decompression could help improve their neurological status.

Keywords: Intervertebral disk extrusion, Thoracolumbar spine, Photobiomodulation, Rehabilitation therapy, Dog

- 針對胸腰椎椎間盤突出(type 1)手術術後，復健配合使用Laser治療預後評估
- 24隻犬分2組，試驗組與對照組，分別為術後復健加Laser，術後復健
- 使用4級Laser術區治療，4 J/cm²每日一次，14天，經30天及60天評估，神經恢復量表呈現神經恢復狀態進步
- 恢復行走能力時間縮短，試驗組14天恢復行走，對照組24天恢復行走

Bruno et al. BMC Veterinary Research (2020) 16:120, https://doi.org/10.1186/s12917-020-02333-3

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t-PEMF 標地式脈衝式電磁場治療

ORIGINAL STUDIES

Effect of Targeted Pulsed Electromagnetic Field Therapy on Canine Postoperative Hemilaminectomy: A Double-Blind, Randomized, Placebo-Controlled Clinical Trial

Leilani X. Alvarez, DVM, DACVSMR, John McCue, DVM, DACVIM (Neurology), Nathaniel K. Lam, DVM, DACVS, Gulce Askin, MPH, Philip R. Fox, DVM, MS, DACVIM, DECVIM (Cardiology), DACVECC

2019

ABSTRACT

Intervertebral disc disease is one of the leading causes of paralysis in dogs. Pulsed electromagnetic field (PEMF) therapy has been advocated for improving wound healing and pain reduction; however, robust clinical trials are lacking. The present prospective, double-blind, placebo-controlled trial evaluated targeted PEMF therapy administered to 53 client-owned dogs who underwent hemilaminectomy for naturally occurring disk extrusion intervertebral disc disease. The dogs were randomized to receive either targeted PEMF (n = 27) or placebo treatment (n = 28). Wound healing, evaluated by visual analog score and wound evaluation scale, was significantly improved at 6 wk postoperatively in the treatment compared with the control group (*P* = .010 and .023, respectively). Pain medications were administered less frequently in dogs receiving PEMF treatment during the 7 day postoperative period compared with the control treatment group (*P* = .010) with codeine administered 1.8 times more frequently in the control group. No untoward effects were recorded in either treatment group. More frequent evaluation of outcome measures with larger patient numbers, as well as histologic samples, may be useful in future studies. Dogs receiving PEMF therapy following postoperative hemilaminectomy demonstrated improved wound scores at 6 wk and reduced mean number of owner-administered pain medications compared with the control group therapy. (*J Am Anim Hosp Assoc* 2019; 55:83–91. DOI 10.5326/JAAHA-MS-6798)



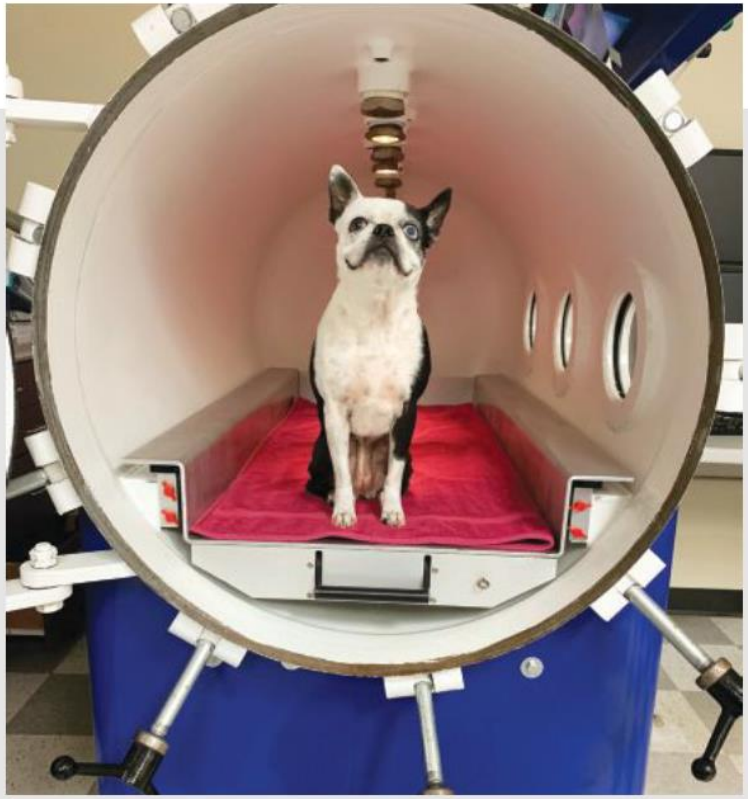
- 針對半椎板切除術後配合使用標地式脈衝磁波圈治療
- 53隻犬分2組，試驗組與對照組
- 住院期間每次15分鐘，一天4次，出院後繼續居家使用，每次15分鐘，早晚各一次，持續1週
- 6週後評估，術區傷口明顯改善較快，且止痛藥使用減少

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HBOT 高壓氧治療

- 高壓氧治療具有神經保護作用
  - 減少氧化壓力
  - 促進血管生成
  - 調控發炎
  - 減少神經系統水腫
- 減緩神經病變痛
- 脊髓損傷修復
- 周邊神經再生
- 降低創傷性腦血管損傷
- 延緩運動神經元疾病發生

- 常用壓力艙
- 1.4 ATA
- 2.0 ATA



Canine patient in the hyperbaric oxygen chamber preparing for treatment.

todaysveterinarypractice.com
JANUARY/FE BRUARY 2020, p69-72

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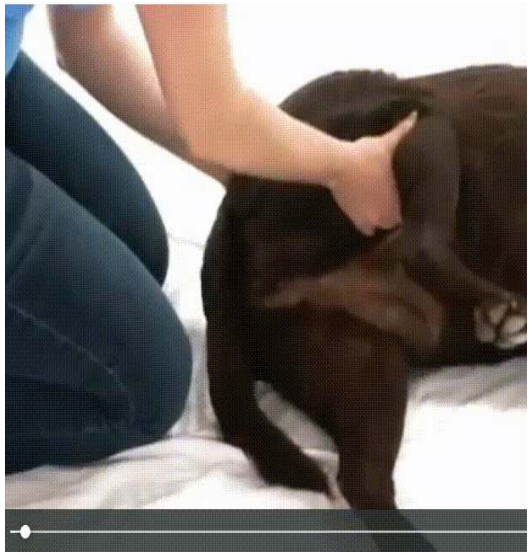
肢體復健方案

- 肌肉與關節及本體感覺
  - PROM 被動關節活動
    - 10-15 rep*3-5/day
  - Massage 按摩
    - 15’ *3/day
  - Limb stretching 伸展
    - 15-20 ‘ *3*3/day
  - Toe-grip增加抓力
  - Assisted standing 協助站立
    - 2-5’ *3-5/day
  - Assisted ambulation 協助行走
    - 5-15’ *3-5/day
- 術後第7天，UWTM 2-3‘ /day，採低速高水位，之後視動物情復原況逐漸增加次數及阻力



BSAVA Manual of Canine and Feline Neurology, 4th Edition







- **PNF (Proprioceptive Neuromuscular Facilitation)**

  本體感覺神經促進術
  - Tapping 拍、敲
  - Quick brushing 擦、刷
  - Quick stretching 拉、伸
  - Tickling paw 搔爪
  - Joint compression 關節按壓
  - Toe pinching with withdrawal reflex 掐腳趾+回縮反射
  - Tail stimulation (traction, massage, pinch) 尾巴刺激
  - Rhythmic stabilization 節律性穩定

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Neuro-inhibitory Techniques

神經抑制(放鬆)術

- Effleurage 輕撫法
- Prolonged slow stretch 伸展拉伸
- Light joint compressions/approximations 關節擠壓
- Cryotherapy 冷療
- Deep tendon pressure 肌腱深按
- Rocking/oscillations 搖擺震盪
- Digit flexion 腳趾屈曲
- Kinesiology tape 肌內效貼紮

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運動復健

- 運動復健目標
  - 關節
  - 本體感覺
  - 肌肉
  - 平衡
- 運動復健項目
  - 跨越橫桿
  - 復健球
  - 平衡板
  - 協助行走
- 居家運動復健
  - 坐站練習
  - 重心偏移
  - 關節活動





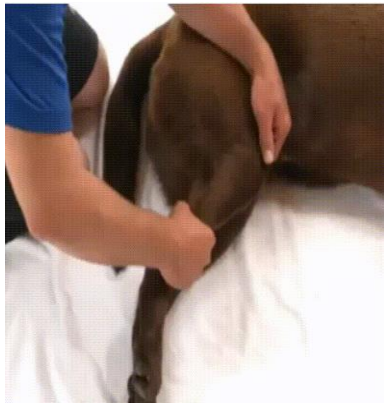












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https://canineexercise.com/

輔具使用

- A-Frame
- Hoist
- Help Em Up
- Sling
- Cart (wheel chair)
- Toe grip
- Toe Up
- Booties
- Theraband











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From Canine Exercise Solutions (https://www.canineexercise.com/); BSAVA congress; and google rearch

居家環境調整













https://canineexercise.com/ ; google.com

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甚麼是退化脊髓病(漸凍症)
Degenerative Myelopathy

- 退化脊髓病是一種運動神經元疾病造成後軀逐漸癱瘓，不痛的脊髓疾病，病程進展緩慢且不可逆。
- 據研究，此與SOD-1 (superoxide dismutase)基因突變有關
- 這種基因突變引起脊髓白質的髓鞘脫失(demyelination)，軸突退化(axonal degeneration)，腰椎脊髓背神經根毀損，甚至最新研究顯示影響腦部紅核(red nucleus)影響步態產生(gait generation)

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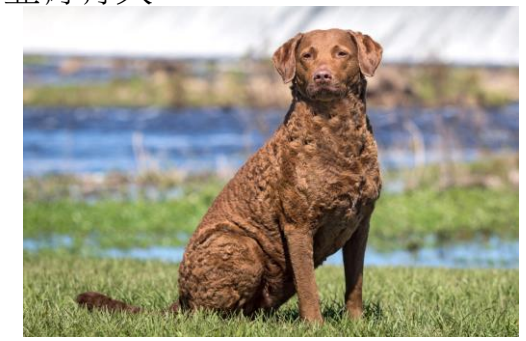
- 好發品種
  - 台灣常見
    - German Shepherd
    - Pembroke Welsh Corgis
    - Boxer 潘布鲁克威尔士柯基犬
    - Wire Fox Terrier 剛毛獵狐梗
  - 國外常見
    - Pembroke Welsh Corgis
    - Boxer
    - Chesapeake Bay Retrievers 乞沙比克海灣獵犬
    - German Shepherd
    - Rhodesian Ridgebacks 羅得西亞背脊犬
    - Bernese Mountain Dogs 伯恩山犬











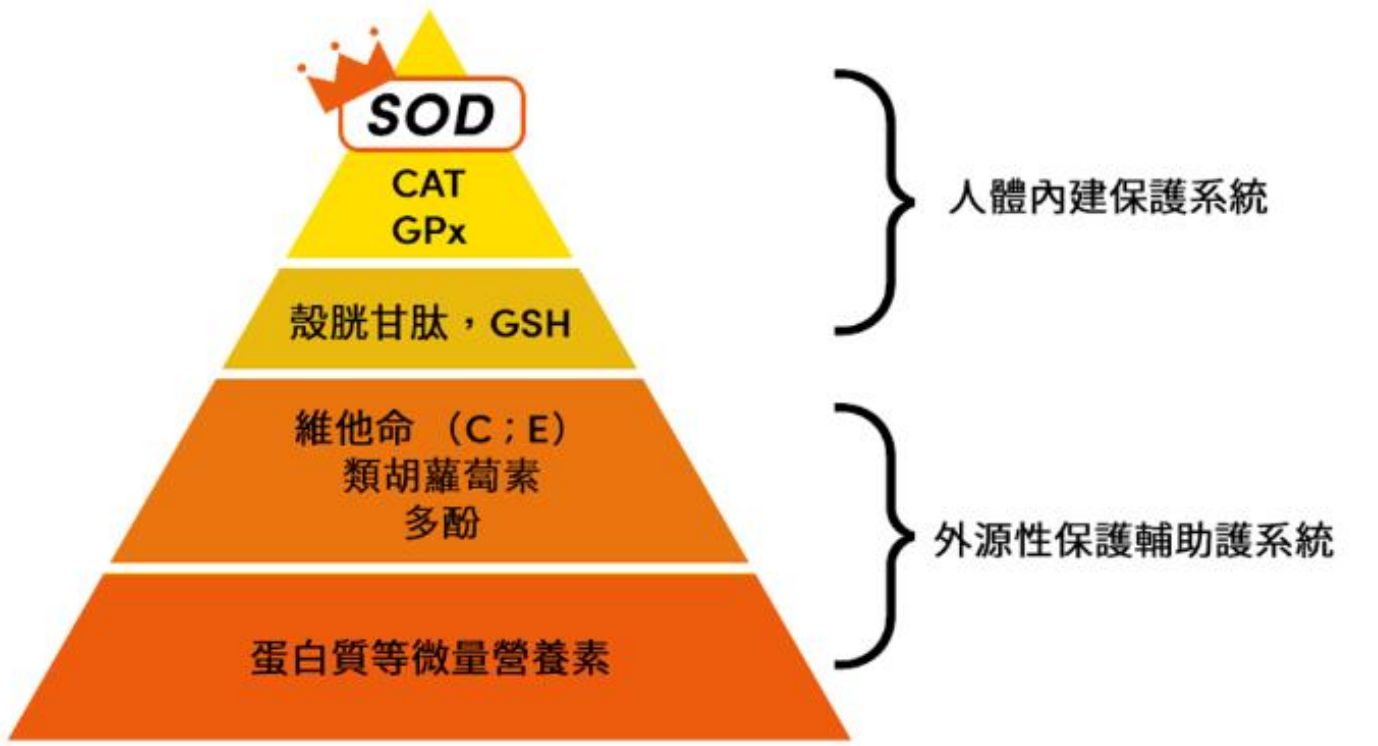




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退化脊髓病診斷

- Clinical signs
- Gene mutation testing
  - SOD1 mutation
  - A/A- at risk, N/A- carrier
- MRI/CSF
- Differential
  - IVDD
  - Tumor
  - Inflammatory
- Definite diagnosis- histopathology



Nadine Pomarede MD, Meera Chandramouli. Enhancing the Skin's Natural Antioxidant Enzyme System by the Supplementation or Upregulation of Superoxide Dismutase, Catalase, and Glutathione Peroxidase. Nutritional Cosmetics. 2009, Pages 245-265.

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寵物種別：	犬	寵物性別：	公
檢體形式：	血液		

檢驗套組 / 檢測類別	正常	異常
退化性脊椎神經病變		●

備註：此次檢測結果為突變基因型(A/A)。若帶有同合子突變基因型(突變型, A/A)有較高罹患風險，而帶有雜合子突變基因型(攜帶型, N/A)則罹患風險較低。

檢測日期：2022-11-15

寵物種別：	犬	寵物性別：	母
檢體形式：	血液		

檢驗套組 / 檢測類別	正常	異常
退化性脊椎神經病變		●

備註：此次檢測結果為突變基因型(N/A)。若帶有同合子突變基因型(突變型, A/A)有較高罹患風險，而帶有雜合子突變基因型(攜帶型, N/A)則罹患風險較低。

檢測日期：2022-12-14

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- 分級
- Stage-1
- 後肢無力，走路步態改變，尾巴位置改變，後肢磨地，後趾甲磨損不均，起立困難
- Stage-2
- 站立困難，後肢窄距，站立時後肢搖晃，腳背著地頻率增加，後肢交叉打架，有可能排糞失禁尤其在夜晚睡覺時
- Stage-3
- 尾巴沒活動，站立或走路經常跌倒，跨越伸肌反射(cross extensor reflex)異常，後肢失去自主運動，糞尿失禁，完全需要協助起身

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上位神經元後軀輕癱及本體感覺共濟失調

Disease Onset

↓ 6-12 mos

UMN Paraparesis and GP ataxia
- Progressive general proprioceptive ataxia
- Asymmetric and spastic paraparesis
- Postural reaction deficits in PL
- Intact spinal reflexes (patellar reflex may be decreased)
- Lack of paraspinal hyperesthesia

Diagnostics – Early
- Normal neuroimaging
- Normal electrodiagnostic testing
- Normal CSF analysis
- Homozygosity for SOD1:c.118G>A

下位神經元後軀輕癱至後肢截癱

↓ 9-18 mos

LMN Paraparesis to Paraplegia
- Mild to moderate loss of muscle mass in PL
- Reduced to absent spinal reflexes in PL
- Nonambulatory paraparesis to paraplegia
- +/- urinary and fecal incontinence

下位神經元後肢截癱至前肢無力

↓ 14-24 mos

LMN Paraplegia to TL Weakness
- Signs of TL weakness
- Flaccid paraplegia
- Absence of spinal reflexes in PL
- Severe loss of muscle mass in PL
- Urinary and fecal incontinence

Diagnostics – Later
- EMG abnormalities
- Nerve conduction studies show temporal dispersion and slow velocities

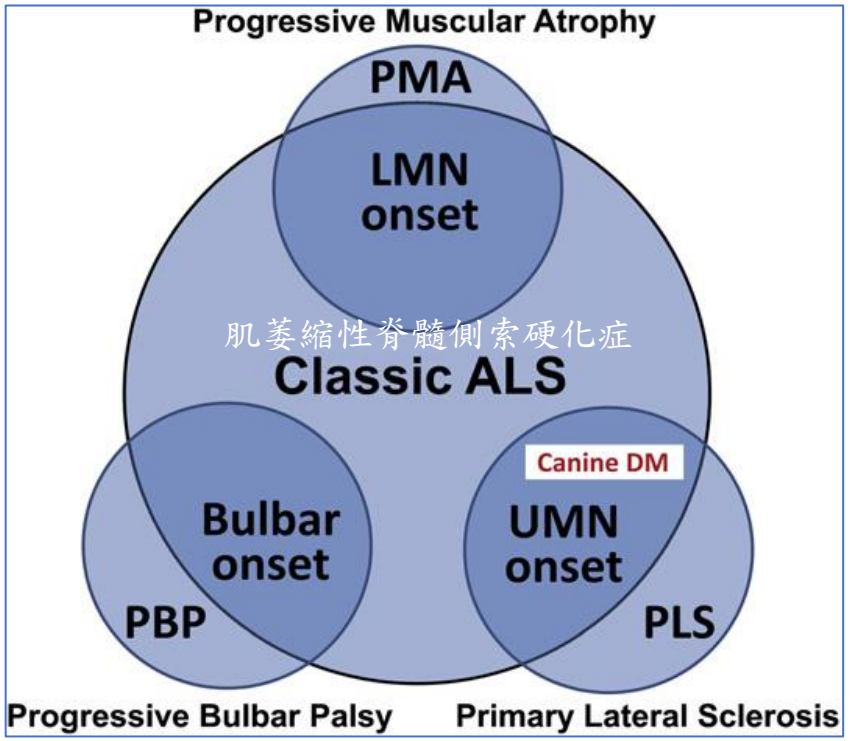
下位神經元四肢截癱及腦幹症狀

↓

End Stage
(> 36 mos)

LMN Tetraplegia and Brain Stem Signs
- Flaccid tetraplegia
- Difficulty with swallowing and tongue movements
- Absence of spinal reflexes all limbs
- Reduced to absent cutaneous trunci reflex
- Generalized and severe loss of muscle mass
- Urinary and fecal incontinence

進行性肌萎縮



進行性延髓麻痺　　原發性側索硬化

Canine degenerative myelopathy, Vet Clin Small Anim 40 (2010) 929–950.

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退化脊髓病治療與照顧

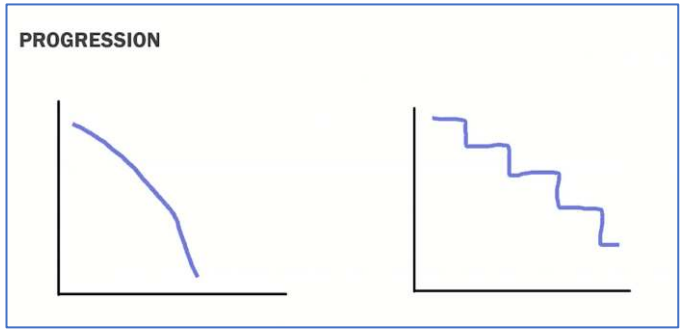
- Acupuncture
- Laser
- PROM
- Massage
- Therapeutic exercises
- Pool/UWTM
- Assisted devices

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Table 1. Prescribed physiotherapy for dogs with degenerative myelopathy (DM).

	Instructions	Duration and frequency
Active exercise 主動運動	• Slow walking • If needed, knuckling is prevented by a sling around the paw and pulling the dog's limb by each step with it • Frequent exercises preferred to long exercises • Exercise has to be adapted to animal's condition • Dog sits and gets up several times • Assistance with a sling if needed • Attention is paid to correct placement of the paws • Weight shifting while standing: making the dog bear his weight once on the left, then on the right side by pushing him gently at the level of the hip • Changing of ground (grass, asphalt, sand) • Stair climbing, walking uphill	5–10 minutes at least 5 times/d
Passive exercise 被動運動	• Gentle, slow extension and flexion of each joint of both hind limbs (starting distally, manipulating of each joint performed separately) • Maintaining the range of physiologic motion of each joint • The limb is always fixed proximally to the joint and the distal part is moved	3 times/d, 10 times in each articulation
Massage 按摩	• Massage is started and finished with stroking • Gentle massage (kneading) of the entire paravertebral muscles and the limbs, starting from distal to proximal	3 times/d
Hydrotherapy 水療	• If available, walking on underwater treadmill; otherwise, swimming or walking in water, depending on dog's ability • Adaptation to animal's condition is important • Assistance with a sling as needed • Weight shifting while standing in water: making the dog bear its weight once on the left, then on the right by pushing him gently at the level of the hip	At least once a week, 5–20 minutes
Paw protection 腳掌保護	• With a bandage or socks and shoes	While walking

疾病進展



√

Daily Controlled Physiotherapy Increases Survival Time in Dogs with Suspected Degenerative Myelopathy, J Vet Intern Med 2006;20:927–932.

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游泳與水中跑步差別

- 游泳
  - 不負重運動
  - 增進自主關節活動
  - 對上半身有益
  - 增進核心及軀幹鍛鍊
  - 有趣



- 水中跑步
  - 可調整式負重運動
  - 增進自主關節活動
  - 本體感覺訓練
  - 增進平衡
  - 術後早期運動



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- 腳背反摺翻正
- 保護及止滑
- 輔具協助











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- Onset of clinical signs
  - > 5 years
  - 平均 9 years in large dog breeds
  - 大多數 > 8 years
  - Corgi average 11 years
- 預後不佳
- 治療目標延長生命至少6個月
- Kathmann 2006
  - 存活(survival time) 55天-完全沒做復健
  - 存活(survival time) 130天-中度做復健(居家運動復健每天3次，水療每週1次)
  - 存活(survival time) 255天-積極做復健（居家運動復健每天3-5次，每天3次按摩及PROM或每天水療）

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Mary Ann Liebert, Inc.
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Photobiomodulation—Review

Retrospective Observational Study and Analysis of Two Different Photobiomodulation Therapy Protocols Combined with Rehabilitation Therapy as Therapeutic Interventions for Canine Degenerative Myelopathy

Lisa A. Miller, DVM, CCRT, CVA,[1] Debbie (Gross) Torraca, DPT, MSPT, OCS, CCRP,[2] and Luis De Taboada, MSEE[1]

2020

Abstract

Objective: The objective of this retrospective review was to examine the impact that adding photobiomodulation therapy (PBMt) to rehabilitation therapy had on the pathology of degenerative myelopathy (DM) in canine patients.

Background: Canine DM is a progressive, fatal neurodegenerative disease for which there exists a dearth of effective treatments, limiting clinicians to pursue symptom palliation.

Methods: Clinical records of dogs referred for presumed DM to a specialty rehabilitation facility were screened for patients meeting study criteria. Qualifying patients were divided into two groups: Protocol A (PTCL-A) and Protocol B (PTCL-B) group, based on the PBMt protocol used. Data related to demographics, diagnostics, rehabilitation protocols, and progression of clinical signs were collected. Data were analyzed to determine differences in outcomes between the two treated groups and historical data expectations, as given by a previously published study.

Results: The times between symptom onset and euthanasia of dogs in the PTCL-B group: 38.2 ± 14.67 months (mean ± SD), were significantly longer than those of dogs in the PTCL-A group: 11.09 ± 2.68 months. Similarly, the times between symptom onset and nonambulatory paresis (NAP) or paralysis of dogs in the PTCL-B group: 31.76 ± 12.53 months, were significantly longer than those of dogs in the PTCL-A group: 8.79 ± 1.60 months. Further, Kaplan–Meier survival analysis showed that the times from symptom onset to NAP of dogs in the PTCL-B group were significantly longer than those of dogs in the PTCL-A group (Mantel-Cox Log Rank statistic = 20.434, $p < 0.05$) or the historical data group (Mantel-Cox Log Rank statistic = 16.334, $p < 0.05$).

Conclusions: The data reviewed show significantly slower disease progression—longer survival times—for patients in the PTCL-B group than those in the PTCL-A group or published historical data. Further studies are warranted.

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- 回溯性研究，不同雷射治療條件治療犬退行性脊髓病(Degenerative Myelopathy)之效果
- A組方案(n=6):
- 復健治療(包括醫院及家中運動復健)+雷射治療， 3級雷射，參數為904nm波長，0.5W，以點對點方式照射，8 J/cm^2 /點，20點，治療面積為650-1000 cm^2, 時間為5分鐘20秒
- B組方案(n=14):
- 復健治療(包括醫院及家中運動復健)+雷射治療， 4級雷射，參數為980nm波長，6-12W，以掃描方式照射，14-21 J/cm^2 ，治療面積為650-1000 cm^2 ，時間為25-26分鐘15秒
- 結果：
- B組由症狀發生至最終決定安樂死時間為38.2個月，顯著比A組久，A組為11.09個月
- 由症狀發生至無法負重後軀輕癱，B組治療31.76個月顯著比A組久，A組為8.79個月
- 結論：
- 以B組治療方案顯著可以延緩病程進行，並且有較長存活時間

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